Senior Design Project Report

CSE 499B

Department of ECE



Early Parkinson's Disease Detection from Gait Using an Explainable CNN-Transformer and Early Gait Instability Index

Submitted By

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APPROVAL

Tasfia Samrin (2211249042), Tayeba Hasan (2211963042) and Tasin Koraiza Boishakhi (2211528042) from Electrical and Computer Engineering Department of North South University, have worked on the Senior Design Project titled "Early Parkinson's Disease Detection from Gait Using an Explainable CNN-Transformer and Early Gait Instability Index" under the supervision of Dr. Shahnewaz Siddique partial fulfillment of the requirement for the degree of Bachelors of Science in Engineering and has been accepted as satisfactory.

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DECLARATION

This is to declare that this project is our original work. No part of this work has been submitted elsewhere partially or fully for the award of any other degree or diploma. All project related information will remain confidential and shall not be disclosed without the formal consent of the project supervisor. Relevant previous works presented in this report have been properly acknowledged and cited. The plagiarism policy, as stated by the supervisor, has been maintained.

The project titled "Early Parkinson's Disease Detection from Gait Using an Explainable CNN-Transformer and Early Gait Instability Index" was developed using the publicly available WearGait-PD dataset. The EGI feature pipeline, all machine learning and deep learning architectures (SVM, 1D-CNN, CNN-Transformer, ResNet-18 with Recurrence Plots), SHAP explainability analysis, and the content of this report are entirely the original contributions of the undersigned students, completed under the supervision of the project advisor.

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Furthermore, we would like to thank the Department of Electrical and Computer Engineering, North South University, Bangladesh for facilitating the research. We would also like to thank our family members, friends, and fellow classmates, to whom we might have caused any inconvenience during the project term, for their understanding and support.

ABSTRACT

Early Parkinson's Disease Detection from Gait Using an Explainable CNN-Transformer and Early Gait Instability Index

Parkinson's Disease (PD) is a progressive neurodegenerative condition that manifests with characteristic gait abnormalities, including decreased stride variability, asymmetry, resting tremor, and impaired turning. As PD symptoms are predominantly motor in nature, early-stage identification remains clinically challenging. This project introduces an automated, objective framework for early PD detection through an Early Gait Instability Index (EGI) using wearable inertial measurement unit (IMU) sensors placed at the ankles.

The EGI derives five composite scores — variability, asymmetry, tremor, timing, and turning from the IMU recordings in the WearGait-PD dataset during free-paced walking and Timed Up-and-Go (TUG) activities, providing clinicians with interpretable, quantitative gait assessments. In addition to the EGI based SVM model, multiple deep learning models were developed and evaluated. The SVM classifier was trained on EGI features and demonstrated stronger predictive performance compared to a baseline classifier without EGI-derived features. Three deep learning models were also implemented: a 1D Residual CNN with Squeeze-and-Excitation (SE) attention, a Hybrid CNN-Transformer, and a ResNet-18 trained on 2D Recurrence Plot (RP) gait images. All models were evaluated using subject-level group 5-fold cross-validation to prevent data leakage and reduce bias. SHAP explainability analysis was applied to validate and interpret the final results. The Hybrid CNN-Transformer achieved the best overall performance, attaining an AUC of 0.8544, accuracy of 79.87%, sensitivity of 80.88%, and specificity of 79.01%.

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Chapter 1 Introduction

1.1 Background and Motivation

Parkinson's Disease (PD) is the second most prevalent neurodegenerative disorder in the world with over 10 million people worldwide affected [1]. It results from the progressive loss of dopamine-producing cells in the substantia nigra of the brain and causes symptoms of bradykinesia, rigidity, resting tremor and postural instability [2]. These symptoms create significant impairments in mobility, balance and quality of life.

Gait problems are among the most frequent symptoms of Parkinson's. Patients may have a slow gait, a small step length, a poor turning ability, freezing of gait, and a loss of balance [3]. These abnormalities make people more likely to fall and be unable to walk around on their own. Gait-related symptoms often occur in early disease and gait analysis is a key research field for the early detection of Parkinson's Disease.

The diagnosis of Parkinson's disease is largely based on clinical diagnosis and neurological examination. Early symptoms are, however, often subtle and hard to be accurately diagnosed [4]. Hence, researchers have been increasingly interested in wearable sensor technologies and in the use of artificial intelligence techniques for objective and automatic evaluation of Parkinson's disease.

Portable, low-cost, and capable of continuous data acquisition of motion signals like acceleration and angular velocity, Wearable Inertial Measurement Unit (IMU) sensors are widely used in gait analysis [5]. IMU-based systems can be used to automatically detect gait abnormalities linked to Parkinson's disease in combination with machine learning and deep learning algorithms.

This project aims to develop an explainable IMU-based gait classification model for detecting Parkinson's disease in the early stages with sensors placed on both of the ankles. The proposed system examines both the clinically engineered Explainable Gait Index (EGI) features and deep learning models like 1D-CNN, CNN-Transformer and ResNet architectures. In addition, explainability techniques like SHAP analysis and attention visualization are integrated for enhanced clinical interpretability. Early diagnosis of Parkinson's disease can help facilitate timely intervention, better management of the patient, and better quality of life [6].

1.2 Purpose and Goal of the Project

The main goal of this project is to create a precise and interpretable machine learning system to detect Parkinson's Disease (PD) in its early stages by analyzing gait using wearable sensors. The goal of the system is to automatically categorize Parkinsonian gait patterns from multichannel inertial measurement unit (IMU) signals collected using bilateral ankle-mounted sensors.

Specifically , this project is aimed at the distinction of early-stage PD patients (Hoehn and Yahr stages 1-2) from healthy controls by analysing gait characteristics in spatiotemporal dimension. This is an automatic and objective evaluation of gait abnormalities, which is different from conventional clinical assessments, which are subjective and time consuming.

A new Explainable Gait Index (EGI) is presented which encompasses five clinically interpretable components – variability, asymmetry, tremor, timing and turning – to quantify gait instability. The index is a single composite biomarker that reflects the severity of motor impairment as well as gait irregularity in Parkinson's disease.

The major contributions of this work include the following:

1. Proposing a new Early Gait Instability Index (EGI) : A novel Early Gait Index(EGI) was introduced which quantify gait instability by integrating five clinically interpretable components such as: Variability, Asymmetry, Tremor, Timing and Turning. EGI acts as a unified composite biomarker indicating the level of motor impairment and gait instability in PD patients.
2. Multi-modal gait representation framework: The proposed framework utilizes raw IMU sensor signals along with Explainable Gait Index features to jointly represent gait at both low-level and high-level.
3. Evaluation of several ML & DL algorithms: Traditional machine learning algorithms such as SVM were compared with Deep Learning algorithms including one-dimensional CNN, CNN-Transformer models, and ResNet-18 with 2D Recurrence Plot architectures in this study to perform classification for Parkinson’s Disease.
4. Focus on early stage PD: The research concentrated on the diagnosis of early stage Parkinson’s disease (H and Y Stage1-2) for its greater clinical challenges in detection.

5. Robust Subject-level Group 5-fold Cross-validation: A highly effective subject-level 5-fold cross-validation method was used to avoid any possible data leakage during testing.
6. Model explainability: SHAP-based feature importance and Attention mechanism-based visualization techniques were used to improve model explainability.
7. Bilateral IMU sensors for gait asymmetry and turning evaluation: Bilateral ankle placement is useful for modeling interlimb asymmetry, turn performance, and gait coordination disorders.

1.3 Organization of the Report

This report is configured in a basic manner. The next section of the report (Chapter 2) focuses on what others have learned about PD gait analysis and machine learning methods. Then in Chapter 3 we talk about the dataset that we used, and how we prepared the data for the EGI feature engineering framework, as well as the machine learning and deep learning architectures that we used. Following this, Chapter 4 illustrates the outcome of our experiments, our learning and how our models compare. This project is expected to impact humans, their health and the environment, which are addressed in Chapter 5 of this report. Chapter 6 is on the planning of the project along with the cost of the project. Chapter 7 is devoted to the engineering problems that we had to solve and what we did. Finally Chapter 8 discusses what we learned; what we didn't do; and what we can do now with the PD gait analysis and machine learning approaches.

Chapter 2 Research Literature Review

2.1 Existing Research and Limitations

1. Dataset Foundation (WearGait-PD)

The WearGait-PD open-access dataset used in this project was published in 2026 by Anderson et al. [1]. The dataset contains raw IMU and sensorized insole data from 100 people with PD and 85 age-matched controls, recorded and synchronized to a gait walkway reference system, consisting of 3-DOF acceleration and rotational velocity from 13 body-worn IMUs. All data were linked to clinical assessments based on H and Y scores and MDS-UPDRS, developed with the assistance of the FDA, VA, and Johns Hopkins School of Medicine.

2. Classical Feature-Based Detection (CoP Insole Approach)

Nanayakkara et al. [2] tested center-of-pressure (CoP) features extracted from wearable pressure insoles applied to the same WearGait-PD dataset for PD detection. The authors selected 39 PD and 38 controls and extracted 144 CoP features from four domains which are: positional, dynamic, frequency, and stochastic during the Timed Up and Go (TUG) test. A 23-feature subset showed the best performance with logistic regression, yielding accuracy 0.875 and ROC-AUC 0.921, where dynamic and frequency-domain asymmetry features were most frequently selected.

3. Deep Learning with Image Representation (GAF-CNN)

Deep Convolutional Networks (DCNNs) and Gramian Angular Fields (GAFs) were used by Mostafavi et al. [3] to classify PD gait signals from pressure insoles. Results showed accuracy of 98.6% and true positive rate of 99.2%, with severity estimated from 10-second gait windows achieving R^2 greater than 0.8 for Hoehn and Yahr scores. While the results appear strong, the dataset and evaluation methodology are not fully specified, limiting reproducibility and direct comparison.

4. Hybrid CNN-Transformer for PD Staging (VGRF Approach)

Naimi et al. [4] proposed a hybrid 1D ConvNet-Transformer model to diagnose PD severity using VGRF signals. The proposed architecture combines CNN and Transformer in an end-to-end model, where the CNN extracts relevant local features from the VGRF signal and the Transformer captures long-term spatio-temporal dependencies. The model achieved 87.89% accuracy on the Physionet gait dataset, outperforming SVM-RBF (75.60%), ANN (76.08%), and standalone 1D-CNN (85.30%) baselines. An ablation study confirmed that removing either the CNN or Transformer component reduced performance, showing both are essential.

5. Transformer Architecture (Self-Attention for Sequence Modelling)

Vaswani et al. [5] introduced the Transformer which is a neural network architecture built entirely on self-attention mechanisms. The Transformer outperforms recurrent models in parallelizability and sequence modelling tasks. Multi-head attention enables the model to attend to information from various representation subspaces simultaneously at multiple positions, facilitating richer temporal modelling. This foundational architecture directly underpins our Hybrid CNN-Transformer, where a CLS token aggregates gait information across 50 temporal positions of each 2-second window.

6. Channel Attention (Squeeze-and-Excitation Networks)

Hu et al. [6] introduced Squeeze-and-Excitation (SE) blocks which adaptively recalibrate feature responses across CNN channels. SE blocks explicitly model interdependencies between channels, achieving significant performance improvements at minimal additional computational cost. In this project, SE blocks are integrated into the 1D-CNN and Hybrid CNN-Transformer to dynamically emphasise the most discriminative of the 128 learned gait features per input window which is an architectural improvement over the basic CNN used by Naimi et al. [4] which did not include channel attention.

7. Explainable AI (SHAP Framework)

Lundberg and Lee [7] proposed SHAP (SHapley Additive exPlanations) which is a unified framework for interpreting machine learning predictions. SHAP assigns a significance score to each feature for a specific prediction, addressing the balance between accuracy and

interpretability in complex models. In this project, SHAP KernelExplainer is applied to the trained EGI-SVM model, providing clinical explainability not considered by Naimi et al. [4] or Mostafavi et al. [3], and revealing that gait variability contributes 61.8% of the total model decision weight.

2.2 Limitations of Existing Work

The following observations have been made after a detailed examination of the literature:

(i) Use of pressure insoles or VGRF sensors instead of ankle IMU. Nanayakkara et al. [2] and Mostafavi et al. [3] employed pressure insoles, while Naimi et al. [4] used VGRF foot sensors from the Physionet dataset. Angular velocity, step asymmetry, and rotational dynamics have not been explored in prior work, and to our knowledge, limited study has been conducted using bilateral ankle IMU channels for PD classification on WearGait-PD.

(ii) No clinically interpretable feature framework. Mostafavi et al. [3] used end-to-end deep learning with no clinical interpretation at the feature level. Nanayakkara et al. [2] derived generic CoP statistics without clinical motivation. Naimi et al. [4] processed raw VGRF signals with no clinical signal intention. None grounded their features in specific PD gait markers such as variability, asymmetry, tremor, timing, and turning.

(iii) No explainable AI validation. Although SHAP [7] and Transformer attention [5] are established explainability tools, none of the reviewed studies applied them to validate model decisions against clinical gait knowledge. Naimi et al. [4] conducted only an ablation study without feature-level or attention-level clinical interpretation.

(iv) Evaluation methodology limitations. Mostafavi et al. [3] reported 98.6% accuracy without subject-level Group cross-validation, risking data leakage. Naimi et al. [4] applied 10-fold cross-validation at the walk-segment level rather than the subject level, allowing segments from the same subject to appear in both training and test folds. Nanayakkara et al. [2] used a single 80/20 split without Group K-Fold.

(v) Multi-class severity staging rather than early-stage binary detection. Naimi et al. [4] targeted multi-class H and Y staging (Healthy/Mild/Medium/High) on moderate-to-severe patients (H and Y 2–3). No reviewed study addressed binary PD vs Control detection specifically for early-stage patients (H and Y 1–2), the most clinically critical and least studied group.

(vi) No H and Y stage-stratified analysis. No reviewed study evaluated performance separately for H and Y Stage 1 and Stage 2 patients. Combining all early PD stages hides whether detection works for the mildest patients who need early intervention most.

Table 2.1: Comparison of Related Works

	Nanayakkara et al. [2] (2025)	**Mostafavi et al. [3] (2025)**	**Naimi et al. [4] (2023)**	**Our Work**
Dataset	WearGait-PD	Not explicitly specified (insole pressure data)	Physionet Gait Dataset	WearGait-PD
Sensor Type	Pressure insoles (CoP)	Pressure insoles	VGRF foot sensors	Bilateral ankle IMU
Sensor Location	Foot (insole)	Foot (insole)	Foot (pressure plate)	Bilateral ankle
Subjects	39 PD + 38 Control	Not specified	93 PD + 73 Control	68 PD + 82 Control
PD Stage Focus	All stages mixed	All stages mixed	H and Y 2–3 (moderate–severe)	H and Y 1–2 only (early stage)
H and Y Stratified Analysis	No	No	No	Yes
Feature Engineering	144 CoP features	None (end-to-end)	None (end-to-end)	61 EGI features (5 components)
Models Used	SVM-RBF, LR, RF, KNN, NB	Deep CNN + GAF	Hybrid 1D ConvNet-Transformer	SVM+EGI, 1D-CNN, CNN-Transformer, ResNet-18+RP
Evaluation Method	Single 80/20 subject split	Not specified	10-fold CV (segment level)	Group 5-fold CV (subject level)
	Nanayakkara et al. [2] (2025)	**Mostafavi et al. [3] (2025)**	**Naimi et al. [4] (2023)**	**Our Work**

Subject-Level Evaluation	Partial	Not specified	No	Yes
Best AUC	0.921	Not reported	Not reported	**0.854**
Best Accuracy	87.5%	98.6%	87.89%	**79.9%**
Sensitivity	75.0%	99.2% (TPR)	Not specified	**80.9%**
Specificity	Not reported	98.5% (TNR)	Not specified	**79.0%**
F1-Score	0.857	Not reported	0.87 (weighted avg)	**0.786**
Explainability (XAI)	None	None	Ablation study only	SHAP + Attention maps
Dual XAI Validation	No	No	No	Yes
Early Stage (H and Y 1–2) Focus	No	No	No	Yes

Table 2.1 demonstrates several important differences between the proposed work and existing studies. Unlike previous approaches that primarily relied on pressure insoles or VGRF foot sensors, the proposed system utilises bilateral ankle-mounted IMU sensors capable of capturing acceleration, rotational dynamics, asymmetry, and turning behaviour simultaneously. Furthermore, prior studies either lacked clinically interpretable feature engineering or relied solely on end-to-end deep learning models without explainability mechanisms. In contrast, the proposed framework combines clinically structured EGI features with machine learning architectures and integrates SHAP and attention-based explainability. Most importantly, previous studies commonly used segment-level or non-grouped validation strategies that may introduce subject-level data leakage, whereas this work applies strict Group 5-Fold subject-level cross-validation for more reliable clinical evaluation.

It should be noted that the studies reviewed above were conducted on different datasets with varying sensor modalities, cohort characteristics, disease severity distributions, preprocessing pipelines, and validation strategies. While some studies used pressure insole or VGRF-based signals, others used full-body IMU systems or alternative gait datasets. Furthermore, no prior work was identified that trained and evaluated models under the exact experimental configuration adopted in this project, including the early-stage-only WearGait-PD cohort (H

and Y≤2), bilateral ankle IMU selection, and strict Group 5-Fold subject-level cross-validation protocol. Therefore, the comparison presented in this chapter is intended primarily to highlight methodological trends, architectural differences, explainability approaches, and research gaps rather than to establish direct numerical superiority between models.

Chapter 3 Methodology

3.1 System Design

The proposed system is a dual-branch machine learning pipeline for early-stage Parkinson's Disease prediction using bilateral ankle-mounted Inertia Measurement Unit (IMU) data. The system is designed as an AI-based gait analysis system, capable of predicting gait instabilities that signify early-stage Parkinson's disease (PD). It processes raw gait signals through preprocessing, feature extraction, machine learning and deep learning to generate classification and risk assessment outputs. The overall architecture of the system comprises of the following main components:

1. Data obtained from wearable IMU sensors.
2. Signal preprocessing and segmentation.
3. Feature Extraction for early gait instability index (EGI)
4. Machine learning and deep learning model training
5. Explainable AI analysis
6. Final prediction and performance evaluation

The overall architecture is presented in figure 3.A and the detailed data preprocessing flow is illustrated in figure 3.B.

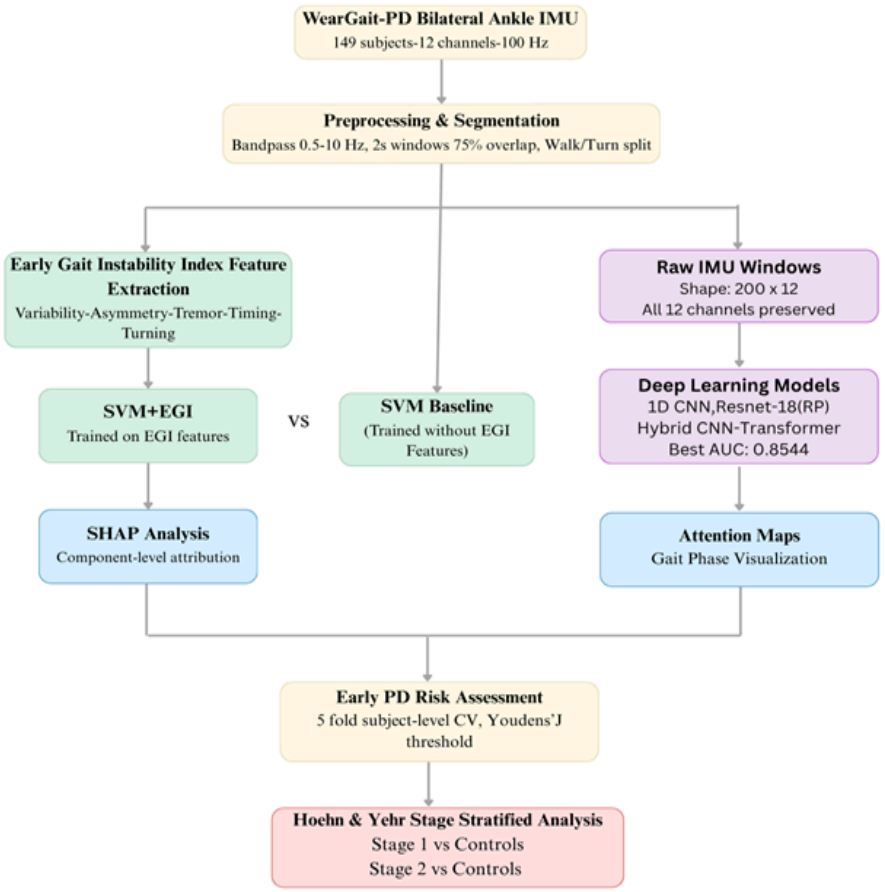


Fig 3.A: The system's full architecture: Two parallel branches.

Figure 3.A shows the full dual-branch system of obtaining raw data and evaluating clinical outcome. The pipeline consists of a shared preprocessing stage, followed by the EGI-SVM branch running on the left and the Deep Learning branch running on the right, both leading to the subject-level evaluation stage, as shown.

The system loads bilateral ankle IMU recordings from 149 subjects, where 82 are healthy controls and 68 early-stage PD patients (Hoehn and Yahr stages 1-2); it is then sampled at 100HZ across 12 channels (3-axis accelerometer and 3-axis gyroscope per ankle). The data is then passed through one common pre-processing pipeline, which later on splits into two separate branches of classifiers.

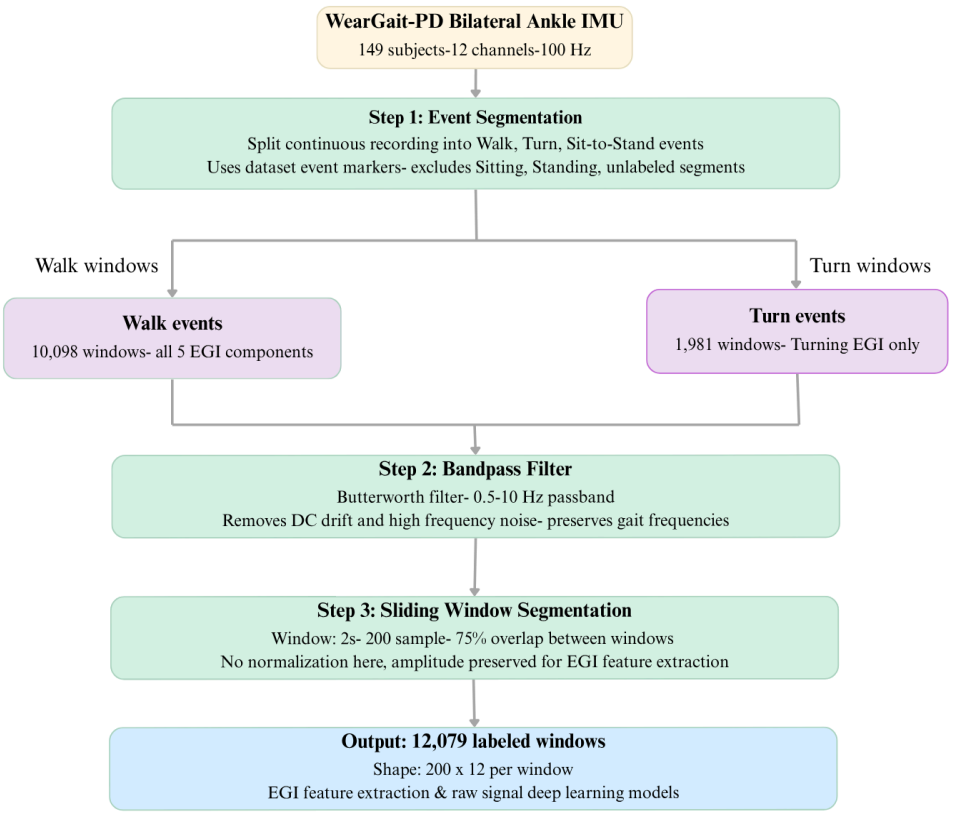


Fig 3.B: The Data Preprocessing Pipeline

Figure 3.B illustrates the step by step data preprocessing pipeline. The common pre-processing pipeline goes through three consecutive steps. The first step is to partition the continuous recording into Walk windows and Turn windows based on the annotated event labels, resulting in only purposeful locomotion events included and non-gait transitions and other events excluded from any window.. In the second stage, a per-channel 4th order zero-phase Butterworth bandpass filter (0.5-10 HZ) filters both the DC drift and high frequency noise per channel while retaining the gait frequency band (0.5-10 Hz) and the pathological resting tremor band (3-8 Hz). This produces a common input of 12079 windows of size 200 x 12 with 75% overlap is provided to both branches as windows that are overlapping and labelled. Once the framework is preprocessed, it splits into two branches as depicted in Figure 3.A.

In the proposed EGI (Early Gait Instability Index) feature extraction module window-shaped gait parameters for each are extracted as five clinically relevant component scores, namely

Variability, Asymmetry, Tremor, Timing and Turning, producing a 55-dimensional structured feature vector for every window. This feature matrix is used to train a Support Vector Machine (SVM) Classifier. Without structuring of EGI, eighty-four baseline raw IMU statistical and spectral features are used to train a parallel SVM baseline as a no-EGI baseline comparison to support the claim of discriminative value added by EGI. To get component-level attribution and generate clinical interpretation, the EGI-SVM model is analyzed using post-hoc SHAP (SHapley Additive exPlanations).

The first part deals with the extraction of Engineered Early Gait Instability features and application of classical machine learning techniques such as Support Vector Machine(SVM). The second branch uses raw IMU signal windows directly as input to deep learning models such as 1D CNN, Hybrid CNN-Transformer and ResNet-based models.

For the deep learning architectures, the raw 200×12 IMU windows are fed directly into the models without any hand-crafted feature engineering. The three models are as follows: 1) 1D-CNN model with SE (Squeeze-and-Excitation) channel attention and residual blocks; 2)Hybrid CNN-Transformer model that replaces Transformer layers with pre-layer norm transformations and includes a classification mechanism for the CLS (Classification) token; and 3) ResNet-18 model applied on 96×96 Recurrence Plot (RP) image representations of the IMU signals. The information provided from each event in the gait cycle for each of the Hybrid model's temporal regions is also extracted via attention map analysis to help determine the temporal regions of the gait cycle that are the most informative for classification.

Both branches are evaluated at the subject level in a shared framework of 5-fold Cross-Validation ensuring that no subject is present in both training and test sets at the same time. For each model a classification threshold is determined optimally using Youden's J statistic. A model based Hoehn & Yahr stage-stratified analysis ensures to re-evaluate the best-performing model separately on H and Y Stage 1 and Stage 2 subgroups against healthy controls, and to measure clinical utility at the earliest detectable disease stages.

3.2 Software Components and Implementation

A. Dataset

The WearGait-PD dataset (Anderson et al., Scientific Data,2025) is obtained from the Synapse open access repository, https://www.synapse.org/Synapse:syn52540892/wiki/. The entire dataset comprises 185 subjects including 100 patients with PD and 85 healthy controls of similar age. Data was programmatically accessed using the synapseclient library and downloaded using token-based authentication, as well as filtering the subjects prior to downloading by a set of clinical criteria. The following image shows the overview of data workflow for the WearGait-PD dataset; it is collected from Anderson et al., Scientific Data,2025.

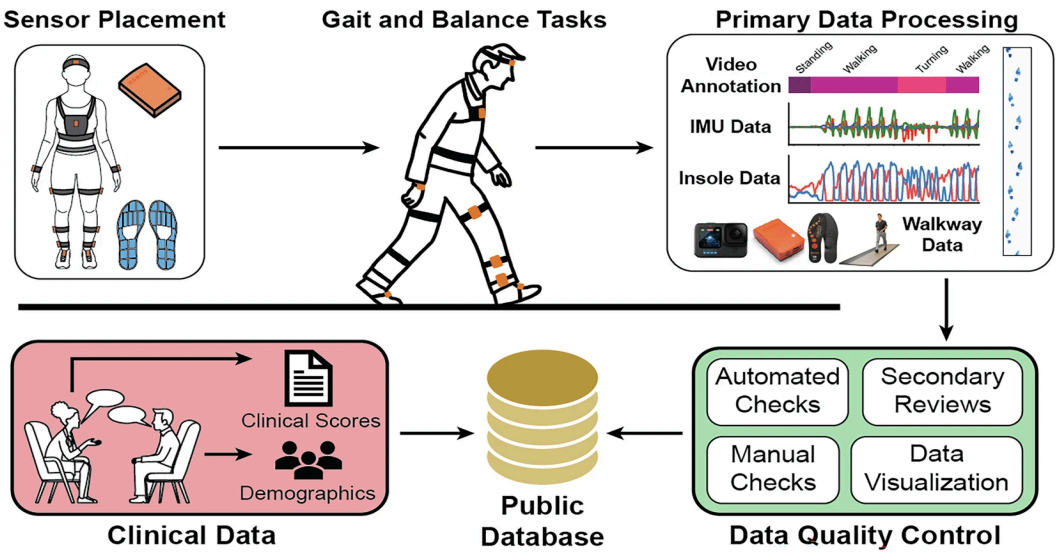


Fig 3.C: WearGait-PD data collection and processing workflow
(https://www.nature.com/articles/s41597-026-06806-2**)**

Subject Selection Rationale:

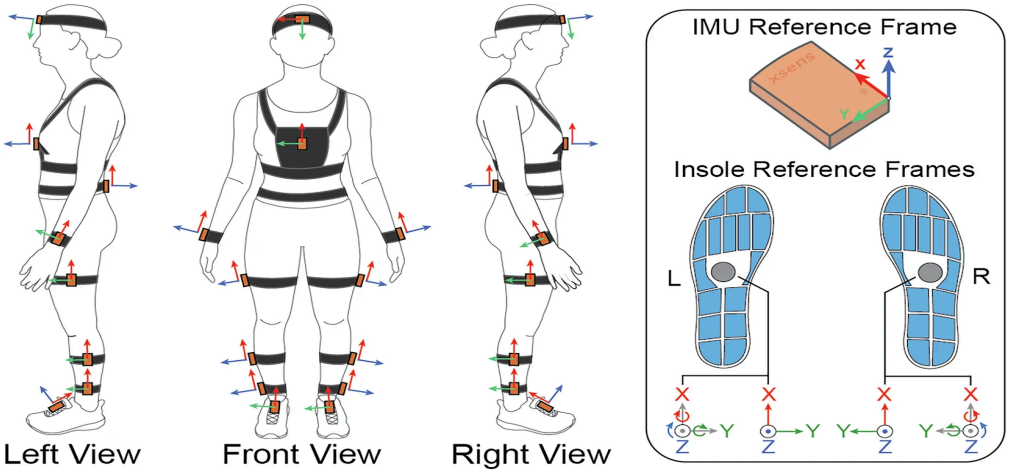
The Modified Hoehn & Yahr (H and Y) scale is a metric that describes PD severity on a scale of 0 (no sign of disease), 1 (disease in 1 hemisphere of the body), 1.5 (disease in 1 hemibody plus axial signs), 2 (disease in both hemibodies, no disability in posture), 2.5 (mild bilateral with recovery of posture), 3 (moderate bilateral with postural instability), 4 (severe, still ambulatory), and 5 (wheelchair-confined). It includes patients from H and Y Stage 1 through to Stage 3 and beyond as the whole data set mean is 2.15 ± 0.61.

This project only covers early stage PD ((H and Y) ≤ 2) -stages 1,1.5,2. The clinical and engineering rationale for this specific cohort selection is threefold: first, patients in early stages exhibit subtle, clinically ambiguous gait deviations that are very difficult to detect and classify using traditional visual assessments. Second, detecting PD at stages 3-5 provides limited clinical value since severe motor dysfunction is already visibly apparent to clinicians without technology and finally, early detection can make a big difference from a clinical standpoint since it will allow drug treatment to begin before further degeneration takes place. After applying the ((H and Y) ≤ 2) filter, 68 PD patients were retained. Of 85 available controls, 82 had complete valid recordings for both task files; 3 were excluded due to missing CSV data. The final cohort is 149 subjects — 68 PD, 82 controls. Each subject contributed two structured gait task recordings:

SelfPace.csv – self-paced walking trial

TUG.csv – Timed Up and Go test (walk, turn, return, sit)

Each recording provides 12 IMU channels at 100 Hz from bilateral ankle sensors: 3-axis accelerometer and 3-axis gyroscope for both left and right ankles. Although the full WearGait-PD sensor suite covers 13 body locations, only bilateral ankle channels are used in this project. This design decision is supported by the biomechanical literature: ankle-mounted IMUs capture the complete gait cycle - heel strike, stance, toe-off, and swing phases - with the highest signal-to-noise ratio for step detection. Ankle sensors also capture tremor propagation to the lower limb and bilateral asymmetry directly from foot contact dynamics.



<u>Fig 3.D</u>: Full WearGait-PD IMU sensor placement (left, front, right views)

B. EDA

EDA was performed before building any feature extraction or model to gain insights about the signal characteristics and confirm that the data is suitable for classification.

Class distribution analysis showed that there was a slight imbalance (55% Control vs 45% PD at the window level). Considering that, this led to the use of inverse-frequency class weighting rather than over sampling in all classifiers, as oversampling time-series windows can lead to introducing temporal artifacts. These differences between PD and Control signals were confirmed by inspection of the raw signal from many subjects:

- The magnitude of acceleration between two gait cycles is greater for PD subjects at the peak of the peak-to-peak range.
- Increase in the power spectral density (PSD) within the tremor band (3-8Hz) is found in PD subjects when compared to controls.
- For PD subjects, the left-right asymmetry for acceleration RMS is greater.

These observations were evaluated using Mann Whitney U and Cohen's d tests. Key findings showed that the Variability component ($d = +0.8$, $p < 0.001$), Asymmetry ($d = +0.6$, $p < 0.001$), Timing ($d = +0.5$, $p < 0.001$) were statistically significant. The tremor component showed marginal but statistically significant separation ($p=0.033$, $d= -0.044$), however the small effect size indicated that it is a weak discriminator during walking. This is physiologically expected since PD resting tremor is partially suppressed during voluntary walking through an action related inhibition mechanism. This EDA finding directly influenced the EGI design ; tremor is retained as a secondary rather than primary discriminator.

A quality check of the signals confirmed no clipped values and demonstrated that the bandpass filtering successfully removed any DC offset. Furthermore, the low amount of missing data, comprising less than 2% of each recording, was effectively handled using linear interpolation.

C. Pre-processing Pipeline

The pre-processing pipeline was engineered in three ways: clinically relevant amplitude of the signals was preserved to extract features, non-gait noise was removed without altering the

frequency bands of the gait and tremor signals, and there was no data leakage into the cross-validation evaluation framework.

A flowchart for data preprocessing pipeline:

Raw CSV Files → Load and check required IMU columns → Linear interpolation of missing values → Butterworth Bandpass Filter → Event label segmentation(Extract walk and turn segments only) → Sliding Window Segmentation → Quality check: reject windows with NaN values → Save: walk_windows, turn_windows.

Missing Value Imputation:

Linear interpolation was used rather than mean/median imputation because the IMU is a continuous time-series with samples that are tightly correlated from one to the next. Local linear trends are used to replace missing samples, which ensures the temporal continuity of the signal, while global mean imputation would cause artificial discontinuities in the signal and then affect the features in the frequency domain.

Bandpass Filtering (0.5–10 Hz):

A 4th order Butterworth filter was chosen for three engineering reasons: the passband of a Butterworth filter is maximally flat, meaning no ripple in the preserved frequency range is generated, unlike Chebyshev filters, the 4th order is sufficiently roll-off to allow the suppression of DC drift below 0.5 Hz and high frequency noise over 10 Hz without introducing excessive phase distortion, and lastly, zero phase filtering through the use forward-backward filter provides zero phase delay, which is critical since a phase shifts would lead to an incorrect alignment of the bilateral left-right signals and corrupt the asymmetry features that rely on precise time-domain synchrony between left and right ankle signals.

The lower cutoff of 0.5 Hz is clinically grounded because it eliminates the DC offset and very slow drift. The upper cutoff of 10Hz includes the normal gait band of 0.5-3Hz, step frequency 0.5-3steps/second and the PD resting tremor band (3-8Hz) with a 2Hz margin, and excludes sensor vibration and electrical noise above 10Hz. Most importantly, at this point, no amplitude normalisation is performed, as amplitude preservation is required because the features used, such as tremor power, gait power, RMS and variability range, are all amplitude dependent. These discriminative features would be lost during premature normalisation.

Event Segmentation:

Walk events labelled Walk and SitToStand and Turn events labelled Turn and TurnToSit are extracted separately. Sitting, Standing and Unlabeled events are disregarded because they contain non gait motion that may introduce label noise in the classification process. Only data sections with a minimum length of 200 samples are retained. Shorter sections are ignored because they cannot form a complete window and may cause edge effects.

Sliding Window Segmentation:

A 2 second window (200 samples at 100 Hz) was selected as it would cover around 1.5-2 gait cycles at a typical cadence of walking (100-120 steps per minute), thus allowing the observation of the kinematic signature of one full gait cycle including stance and swing phases. With a 75% overlap, the number of training windows maximizes the amount of subject data without each window being isolated from the others, thus not losing the temporal continuity of gait patterns. This yields around 13-15 windows per gait segment.

Per-channel z-score normalisation is intentionally performed in the inner loop of the cross-validation, using fit parameters for the training fold and testing with the test fold, in order to avoid any statistics based on test subjects from being used to directly influence the normalisation parameters. This is a crucial implementation choice that avoids a hidden type of data leakage which is often ignored in time series ML pipelines.

D. Early Gait Instability Index Feature Extraction

The Early Gait Instability Index (EGI) is the primary methodological contribution of this project. It is a framework for clinical feature engineering which extracts 61 features from every 200×12 IMU window and groups them into five clinically-motivated components, each of which is directly associated with a documented pathomechanical signature relevant to early-stage Parkinson's Disease.

In contrast, the baseline SVM model uses a collection of handcrafted statistical and signal processing features computed directly from the raw IMU channels and derived magnitude signals. These include channel wise statistical measures (mean, standard deviation, RMS, range, IQR, skewness, and kurtosis), spectral power features, asymmetry measures, and jerk based smoothness descriptors. However, unlike EGI, these features are not structured into

clinically interpretable components linked to specific gait abnormalities associated with Parkinson's Disease. There are three engineering justifications for EGI rather than this raw approach. First, EGI offers a clinical basis: each component represents a known PD gait abnormality thereby maintaining model behavior's interpretability for clinicians. Second, regarding the feature organisation, EGI provides a structured one through grouping the related measurements into clinically coherent components, which allows learning from physiologically meaningful representations instead of just a single statistical descriptor. Third, EGI enables feature importance explainability at the component level and makes it possible to explain feature importance in a clinically relevant manner, where feature sets are not that informative.

Four bilateral magnitude signals are computed from the 12 raw IMU channels using the L2 norm before computing any EGI component. Both Left and Right Acceleration Magnitudes are calculated as follows:

$$L_acc = \sqrt{((ch0^2) + (ch1^2) + (ch2^2))}$$

$$R_acc = \sqrt{((ch6^2) + (ch7^2) + (ch8^2))}$$

and the magnitudes of the bilateral gyroscope as:

$$L_gyr = \sqrt{((ch3^2) + (ch4^2) + (ch5^2))}$$

$$R_gyr = \sqrt{((ch9^2) + (ch10^2) + (ch11^2))}$$

These reduce the three spatial axes to a single signal of locomotion intensity for each ankle. The Z-axis gyroscope channels (L_gyr_z = ch5, R_gyr_z = ch11) are also kept for the Turning component because the vertical axis represents the main turning rotation. These magnitude signals are rotation-invariant and do not require fine-tuning due to small variations in sensor placement orientation across subjects, which is a crucial factor for the successful generalisation of these signals across subjects in clinical IMU datasets.

The 61 features extracted by EGI are grouped into five components: Variability, Asymmetry, Tremor, Timing and Turning, described in detail below.

Component 1- Variability

Dopaminergic malfunction has been correlated with reduced rhythmic gait control, due to basal ganglia dysfunction, which results in an increase in stride to stride variability. This is one of the

earliest and most consistent kinetic features of Parkinson's Disease gait and can be observed at Hoehn and Yahr (H and Y) stage 1 and 2.

The extraction of features for this component involves computation of 11 primary values. The main indicator used for assessment is the Coefficient of Variation (CV), which is computed separately for each accelerometer and gyroscope directions using the formula: $CV = \sigma / (\mu + \varepsilon)$, where $\varepsilon = 1 \times (10^{-10})$ is introduced in order to avoid dividing by zero. Since this value is normalized, its use allows the comparison of variability between patients with different walking speeds and signal amplitudes.

The following statistical descriptors of variability (complementary to max and min) are also calculated for L_acc and R_acc: peak-to-peak range (max- min), interquartile range (P75 - P25), and standard deviation. An addition of 18 directional features are computed by calculating the per- axis mean, standard derivation and RMS for the 6 acceleration channels (L_acc_x/y/z and R_acc_x/y/z), capturing directional gait characteristics that magnitude-based representations alone cannot provide. Together, these constitute the full 29 features of the variability component.

The coefficient of variation varies from limb to limb and the variability score is the mean coefficient of variation in both the limbs:

$$variability_score = (CV_L_acc + CV_R_acc) / 2.$$

Component 2: Asymmetry

PD is typically begins on one side at Stage 1 and then progresses on the same side, creating an ongoing left-right difference in kinematics that can be directly quantified by measuring the kinematics of the two ankles using bilateral sensors, and is the most clinically direct measure of early unilateral PD onset.

Seven features are extracted. The primary measure is the Asymmetry Index, defined as a normalised percentage difference in signal magnitude:

$$100 \times |RMS_L - RMS_R| / (RMS_L + RMS_R + \varepsilon),$$

where, ε is a constant to prevent the denominator from being zero. This normalised formulation eliminates any confounding effect by individual walking speed and effort, and ensures that the index is subject independent and directly comparable over the cohort.

Additional features include the absolute RMS difference (|RMS_L − RMS_R|), the RMS ratio (RMS_L / (RMS_R + ε)), the absolute mean difference (|μ_L − μ_R|) and the absolute standard deviation difference (|σ_L − σ_R|). Furthermore, L-R correlation coefficient (corrcoef(L_acc, R_acc)) quantifies the temporal synchrony of the two limbs; a high L-R correlation is evidence of high temporal synchrony between the movements of both limbs, which is characteristic of a healthy walker, while a low L-R correlation is characteristic of a PD patient, as their movement in one limb is impaired.

Asymmetry index is set to be the component score: asymmetry_score = asym_index.

Component 3: Tremor

PD is also associated with resting tremor in the frequency band 3-6 Hz. Even though voluntary walking suppresses the presence of classical resting tremor through action inhibition mechanisms, subclinical high-frequency oscillatory activity remains in the lower limb while walking. In particular, higher levels of energy across the spectral band from 3 to 10 Hz compared with the normal gait frequency band 0.5–3 Hz yield a measurable signature of PD in the frequency domain, even when walking.

A power spectral density (PSD) is calculated using scipy.signal.periodogram and eight features are extracted. Ankle tremor band power is calculated as the sum of PSD values in the range of 3-10 Hz and gait band power is calculated in the 0.5-3 Hz range for each ankle. Thus, the ratio of the power of the tremor and the power of the gait is then defined as:

$$\text{tremor_ratio} = \text{tremor_power} / (\text{gait_power} + \varepsilon),$$

which is a measure of the increased amplitude of high frequency pathological activity compared to normal gait rhythm.

The Freeze Index (FI), which is a composite measurement, is also calculated as the ratio of the total tremor power to total gait power in both the arms:

$$FI = (\text{tremor_power_L} + \text{tremor_power_R}) / (\text{gait_power_L} + \text{gait_power_R} + \varepsilon)$$

The above equation is derived based on previously described clinical measurements of Freezing of Gait (FOG). Finally, the mean ratio of bilateral tremors is considered as the third component measure: tremor_score = (tremor_ratio_L + tremor_ratio_R) / 2.

Although this component has not yet been adequately separated from other components in the exploratory data analysis (Mann–Whitney U test, $p > 0.05$), it does not perform well on its own

during walking, as would be expected in a physiology of the suppression of resting tremor during voluntary motion. It is complementary, however, in the context of the overall EGI framework, with the provision of spectral information.

Component 4: Timing

The presence of Parkinson's disease disrupts the timing circuits in the basal ganglia, leading to impaired regularity and smoothness in gait, which results in diminished rhythmicity, an increase in the jerky nature of movements, and a broader distribution of spectral power. All of these can be quantitatively assessed using IMU signals. For this aspect, seven features are extracted. Spectral entropy is calculated as $-\Sigma\ p(f) \times \log(p(f) + \varepsilon)$ based on the normalized power spectral density (PSD) of L_acc, where p(f) denotes the normalized distribution of the spectrum. This quantifies how signal power is spread across different frequencies; a healthy gait typically shows concentrated energy at the primary step frequency, leading to lower entropy, while gait in individuals with Parkinson's disease shows a more distributed spectral energy pattern, indicating irregular timing.

Jerk root mean square (jerk_RMS) is the RMS value of the second derivative of L_acc, multiplied by the frequency squared (fs^2). Increased jerk values correspond to decreased smoothness of movements in PD. The smoothness index is defined as 1 / (1 + jerk_RMS), and is bounded for increasingly smooth motion, with higher values indicating smoother motion. Peak Jerk, which is also defined as the maximum absolute second derivative, is used to identify extreme transient movement irregularities.

The regularity of stride is measured by the autocorrelation of L_acc at the first prominent peak after a 10 sample lag. The closer the value is to 1, the more periodic the gait is, and the closer to 0, the more irregular the stepping pattern. The most dominant gait frequency is the one that has the highest PSD in the 0.5-3Hz range.

The component score is computed as a scaled combination of the most discriminative temporal irregularity measures:

$$\text{timing_score} = \text{entropy}/5 + \text{jerk_rms}/1000,$$

in which the scaling factors are to make each term comparable in its numerical range.

Component 5: Turning

Turning subtasks of the Timed UP and Go (TUG) test is particularly impaired in PD. Patients usually show decreased peak angular velocity, multiple-step turning strategy, prolonged turning time, and bilateral gyroscope asymmetry.

This component is computed only within Turn windows; all feature values are set to zero within Walk windows according to the is_turn flag.

Eight features are extracted from the mean bilateral gyroscope value of Z axis, given by:

$gyr_z = (L_gyr_z + R_gyr_z) / 2.$

Four statistical measures such as mean, max, std, and peak-to-peak range of |gyr_z| are used to measure magnitude and variability of turning motion in terms of rotational angle.

Rotational jerk is measured as the root-mean-square, RMS, of the first derivative of gyr_z scaled by the sampling frequency fs, where lower values correspond to smooth turning motion. Rotational smoothness index is given by:

$$turn_smoothness = 1 / (1 + turn_jerk_rms).$$

The difference in bilateral RMS of rotational gyroscope values is a measure of turning asymmetry between the left and right feet during the turning process and calculated as:

$$|RMS(L_gyr_z) - RMS(R_gyr_z)| / (RMS_L + RMS_R + \varepsilon).$$

Component score calculation formula includes four features and is given by:

$$turning_score = 1.5 \times (std/5 + jerk_rms/50 + asymmetry + range/10).$$

These scaling coefficients (1.5, /5, /50, /10) are not learnable parameters; they are normalisation constants selected through experimentation for the purpose of ensuring that all features contribute equally, since otherwise a single feature may predominate within the composite score.

Composite EGI Score

The five scores are then summed up and normalized to form the final composite index of gait instability for each time window:

EGI = (variability_score + asymmetry_score/30 + tremor_score + timing_score + turning_score) / 5.

The asymmetry score is normalized by dividing it by 30 before aggregation, since its unit of measure is percentages (0-100) and the other four components are in the range of 0-1 approximately. Otherwise, the asymmetry component will dominate the aggregation process and make the others irrelevant.

The is_turn flags are removed from the feature set used for SVM training. The is_turn flag is not related to physiological measures but more to data provenance. Therefore, incorporating this flag will be a task-identifier that could potentially induce classification bias in the prediction model. Thus, the SVM is trained using only features derived from the EGI, as opposed to 110 features in the original model.

E. Machine Learning and Deep Learning Models

E1. SVM with EGI- Primary Classical Model

A Support Vector Machine (SVM) with a radial function (RBF) kernel (C=10, gamma= 'scale') was selected as the primary classical classifier. Compared with other models such as Random Forest, Logistic Regression, and k-Nearest Neighbours (k-NN), this algorithm works better with moderate-high dimensional features and few subjects, as margin maximization helps to generalize well and prevent subject-level overfitting.

RBF kernel allows building non-linear decision boundaries without explicit feature expansion. Furthermore, inverse frequency class weight balancing is applied for mild class imbalance correction without oversampling in time series data.

For implementation, sklearn.svm.SVC with probability=True and group 5-fold cross-validation are used. In each fold, missing values are imputed with column-wise median values from the training set. Then, standard scaling with StandardScaler is applied to the train set only to avoid data leakage.

The probabilities obtained from each window for the subject are combined through averaging of all the windows associated with the same subject to get one PD probability measure for each subject.

E2. SVM Baseline- Raw IMU without EGI

To measure the contribution of the proposed EGI architecture, a second SVM model using only raw statistical IMU features without clinical engineering is also trained. The features used for the baseline model include independent channel-wise statistical measures such as the mean, standard deviation, RMS, range, IQR, skewness, and kurtosis. In total, 110 features are extracted per window for the baseline model.

This baseline acts as a benchmark for comparison between clinically engineered structured features and traditionally unstructured statistically extracted features using the same methodology.

E3. 1D CNN with Residual and SE Attention Blocks

A one-dimensional convolutional neural network (1D-CNN) working on the 200×12 raw IMU window signals transposed to 12×200 in accordance with the PyTorch channel-first convention was chosen as the first deep learning model for training due to the fact that a convolutional network will be able to detect the temporal patterns of gaits automatically, without any manual feature selection process.

The architecture involves four residual blocks with increasing filter sizes from 64 to 256, while the kernel sizes increase from 7 to 3. Skip connections (He et al., 2016) facilitate stable deep convolutional networks by enabling gradient flow in the network. After each residual block, there is an SE (Squeeze-and-Excitation) attention mechanism (Hu et al., 2018), which enables channel-wise re-scaling and learning the relevance of the different IMU channels for the network. This is especially important for gaits' analysis, as the ankle accelerometer and gyroscope have distinct significance for locomotion characterization.

Focal Loss, 2.0 is chosen instead of classical cross-entropy because the former gives more weight to hard or ambiguous training samples while down-weighting easy cases to minimize losses due to overfitting.

Gaussian noise injection (σ=0.05), amplitude scaling with a uniform distribution between 0.9 and 1.1, time-shifting +/-10 samples, and channel dropout, 0.1, augmentation techniques are used only in the train phase. They introduce natural variations for a realistic situation but do not affect class labels.

A linear warm-up for 5 epochs is applied before cosine learning rate decay to ensure better initial convergence. Furthermore, there is a 1.6×PD class-weight boost in combination with

frequency-based weights in favour of sensitivity, which is more clinically valuable than specificity. Finally, early stopping with patience=15 is used to get the optimal model in terms of validation-AUC.

E4. Hybrid CNN-Transformer Model

While the 1D-CNN is able to well capture the local temporal gait patterns, it has limited capability for the model of the long-range dependencies over the entire gait sequence. To address such a drawback, a hybrid CNN-Transformer architecture is proposed.

The raw 200×12 IMU sequence is initially processed with a 3-block CNN encoder with SE attention, which cuts the length of the IMU sequence from 200 to 50 and expands the feature dimension from 12 to 128. This enables the Transformer module to learn to understand the higher-level temporal relationships, avoiding the need for low-level signal extraction.

A learnable CLS token is attached to the sequence in front of Transformer encoding. The CLS token is used for self-attention to fuse the global sequence-level information for final classification.

Pre-LayerNorm (Pre-LN) is employed in the Transformer layers to overcome the problems of standard Post-LN, as it has been proven to be better for optimisation stability and gradient propagation for smaller datasets. The final architecture is composed of 4 attention heads and 2 Transformer layers, which is a trade-off of modelling capacity and risk of overfitting.

The final model also features data augmentation, learning-rate warmup, focal loss and extended early stopping for better generalisation.

E5. ResNet-18 with Recurrence Plot Representation

The ResNet-18 model consists of a novel approach to gait classification, which transforms every IMU window into a Recurrence Plot (RP) image and subsequently classifies it as a 2D image classification problem. RPs store temporal gait features such as periodicity, regularity, complexity in a spatial representation which can be analysed by the trained CNN feature detectors. Gaussian-like thresholding has been applied rather than binary threshold to retain graded similarity information.

The bilateral asymmetry of the accelerometers ($|L_acc - R_acc|$) is captured in Channel 1, the average of the bilateral accelerometers $(L_acc + R_acc)/2$ in Channel 2, and the mean of the

Z-axis of the bilateral gyroscope in Channel 3, resulting in each window being transformed into a 3 channel 96×96 RP image.

Because there are not enough windows to train an 18-layer CNN from scratch, the pretrained ImageNet CNN is used for transfer learning. The last fully connected layer is followed by a custom classification head (512→256→BatchNorm→GELU→Dropout→2). By using the Mixed-precision training, which includes torch.cuda.amp.autocast, the memory consumption on the GPU can be lower and the training speed can be faster.

F. Evaluation Framework

Group 5-Fold Cross Validation has been used for all five models. Standard k-fold randomly distributes individual windows among folds, thus enabling that some windows may be present in both training and test sets, even if they come from the same subject. This means that the model ends up learning subject-specific gait signatures instead of generalising PD patterns, which is a severe degree of data leakage that overestimates reported values. GroupKFold puts all windows of a specific subject into only one of the folds, with 100% subject-level separation.

The window-level probabilities are averaged over each test fold to calculate a single subject-level PD probability for each fold, which represents a clinically realistic case where a decision is made on a patient level rather than a 2-second window level. So the optimal decision threshold is selected using the Youden's J statistic (J = Sensitivity + Specificity − 1) on the ROC curve at the subject level, instead of the default threshold of 0.5, the statistically correct method for imbalanced binary medical classification that maximises sensitivity and specificity at the same time. Bootstraps with 1000 resamples are calculated for AUC to measure the estimation uncertainty in the H and Y Stage 1 subgroup (n = 10).

G. Feature Selection, Ablation, and Attention-Based Interpretability

The EGI ablation study removes each of the five component groups one at a time, and the SVM is trained again with the same Group 5-fold cross validation, while the drop in AUC is recorded.The component that leads to the biggest drop in the value of AUC when removed is considered the one carrying maximum information, thereby validating the notion that the constructed groups complement each other discriminatively.

Approximate Shapley values of the SVM probability output are computed using SHAP KernelExplainer with k=100 centroids derived from k-means summarisation of the training set, evaluated on 300 test windows, of which 150 are PD and 150 Control. To represent the training distribution with the summarised distribution without significantly increasing the computational cost, k-means summarisation is used instead of random sampling. The feature-level SHAP values are then combined within each EGI component group to provide component-level attribution, which allows the understanding of the impact of engineered gait mechanisms on classification performance.

Attention based interpretability is also taken out of the hybrid CNN–Transformer model in addition to the post hoc attribution.Besides post hoc explanation, attention based interpretability is extracted from the hybrid CNN–Transformer model for further analysis of the internal representation learning. The CLS token is used to get temporal attention in both the Pre-LayerNorm transformer layers and the CLS token is the one that gathers information from every position in the sequence. The CLS-attention weights are separated in every layer and then averaged across the layers by a geometric mean for Parkinson's and control subjects to gain a stable temporal attention profile throughout the gait cycle.

Temporal attention differences between classes are quantified using a Mann–Whitney U-test at each gait phase, which determines the gait intervals with a statistically significant difference between the classes in their attention allocation. This yields a statistical characterization of discriminative temporal focus learned by the model, as a function of its position.

The Feature-level attention is obtained from Squeeze-and-Excitation (SE) blocks in the CNN encoder that produce channel-wise gating weights on the 128-dimensional feature representation. These weights are the results of adaptive re-calibration of the IMU-derived feature channels. To determine consistently emphasized feature dimensions, mean SE weights are calculated for PD and control groups and compared.

SHAP, ablation and attention mechanisms combine to give a multi-layer interpretability framework. The post-hoc feature and component attribution computed by SHAP, the causal evaluation of engineered feature group importance computed by ablation, and the model's attention mechanisms revealed the model's temporal and feature-level focus. Uniformity of these views enhances the trust in patterns learned from data and increases the interpretability of both engineered and deep learned representations.

H. Clinical Stage-Stratified Evaluation Using Hoehn and Yahr

The Hoehn and Yahr (H and Y) stratification analysis evaluates the best performing model, the Hybrid CNN-Transformer, on clinically meaningful disease stages. The model is tested on H and Y Stage 1 vs Control, Stage 2 vs Control in addition to its performance on the full dataset.

During subject matching, a case sensitivity mismatch between subject IDs in the preprocessed numpy arrays (lowercase) and the clinical CSV (uppercase) was resolved by converting both sides to lowercase before matching, which ensured all 68 PD subjects were correctly linked to their clinical metadata.

H and Y score was then obtained from the clinical metadata and divided into two categories: stage 1, representing unilateral motor dysfunction with minimal functional impairment, and stage 2, representing early bilateral involvement without significant postural impairment.

The model was tested in three experimental settings under identical Group 5-fold cross-validation procedure with regard to the following groups –

1. all patients diagnosed with Parkinson's disease vs. control group,
2. Stage 1 Parkinson's disease vs. control group,
3. Stage 2 Parkinson's disease vs. control group.

All experiments were carried out with the same model architecture, preprocessing pipeline, and hyperparameters configuration. Given that there is only a small number of Stage 1 patients, the bootstrap method with 1,000 samples was used to calculate the 95% confidence intervals of AUC metrics.

The results obtained through this stratified approach showcase the capabilities of the algorithm at various levels of severity of the disease, proving its efficiency in recognizing Parkinson's disease at the earliest stages of development.

Table 3.1: Software Tools used for our project

Tool	Function	Other Similar Tools	Why Selected
Python 3.10	Primary programming language for all pipeline stages	R, MATLAB, Julia	Dominant language in ML/DL research
NumPy / SciPy	Signal processing, bandpass filtering, windowing, statistical tests, periodogram, RP generation	MATLAB Signal Toolbox	Fast vectorised operations; signal.butter, filtfilt, periodogram built-in
Pandas	Data loading, feature dataframe construction, EDA	Polars, Dask	A general table tool that can be directly integrated with scikit-learn
Scikit-learn	SVM, GroupKFold CV, StandardScaler, ROC/AUC metrics	Weka, RapidMiner	Only major ML library which has native implementation of GroupKFold for subject-level leakage prevention.
PyTorch 2.x	1D-CNN, Hybrid CNN-Transformer, ResNet-18 training	TensorFlow/Keras, JAX	Dynamic computation graph enables flexible custom architecture design
TorchVision	Pretrained ResNet-18 with ImageNet weights	timm	Official PyTorch library; standardised pretrained model loading

Tool	Function	Other Similar Tools	Why Selected
Python 3.10	Primary programming language for all pipeline stages	R, MATLAB, Julia	Dominant language in ML/DL research
SHAP	KernelExplainer for SVM component-level attribution	LIME, Captum	Model-agnostic, theoretically grounded Shapley values, built-in visualisation
Matplotlib / Seaborn	ROC curves, confusion matrices, box plots, radar charts, attention heatmaps	Plotly, Bokeh	Static figures without moving parts and direct integration with NumPy;
Kaggle Notebooks (GPU)	Cloud training with NVIDIA Tesla P100 16GB	Google Colab, local CUDA	Free GPU quota, persistent output storage, native dataset integration
synapseclient SDK	Authenticated programmatic download filtered by H and Y ≤ 2.0	Direct browser download	Reproducible automated retrieval without manual file selection

Chapter 4 Investigation/Experiment, Result, Analysis and Discussion

4.1 Dataset Overview and Window Extraction Results

The raw IMU recordings were divided into overlapping 2-second windows containing 200 samples at 100 Hz, with a 75% overlap, or a step of 50 samples. Walking and turning events were extracted separately using the event labels in the dataset, resulting in a total of 12,079 windows across all subjects.

Table 4.1: Preprocessing Output

Data Type	Shape	Control Windows	PD Windows	Total (PD window + control window
Walk	(10098, 200, 12)	6017	4081	10,098
Turn	(1981, 200, 12)	1228	753	1,981
Total	(12079, 200, 12)	7245	4834	**12,079**

Here, the shape represents 9windows x sample x channel), 12 IMU channels per window (3-axis accelerometer + 3-axis gyroscope from both left and right ankles), a total of 12,079 windows were generated across all individuals. Each window spans 2 seconds, 200 samples at 100 Hz. Due to the disparity in subject count (82 controls vs. 68 PD), the class distribution is unbalanced, with 7,245 control windows (59.9%) and 4,834 PD windows (40.1%). Inverse-frequency class weighting was used in all models to address this imbalance.

4.2 EGI Component Score Analysis

Each of the five EGI component scores was statistically assessed between the PD and healthy control groups using Cohen's d effect size and the Mann-Whitney U test across all 12,079 windows following the extraction of 61 characteristics per window using the Early-Stage Gait Index (EGI) framework.

Table 4.2: EGI Component Score Distributions: PD vs. Control (All Windows)

Component	Control (mean ± std)	PD (mean ± std)	Cohen's d	p-value	Sig.
variability_score	0.9334 ± 0.171	0.8954 ± 0.154	−0.233	0.0000	***
asymmetry_score	10.3829 ± 10.565	9.1294 ± 9.128	−0.127	0.0000	***
tremor_score	0.5257 ± 0.173	0.5180 ± 0.177	−0.044	0.0330	*
timing_score	7.0136 ± 2.976	7.9041 ± 3.496	+0.274	0.0000	***
turning_score (turn windows only)	1.2184 ± 0.424	1.2492 ± 0.427	+0.072	0.0248	*
EGI (composite)	1.8051 ± 0.566	1.9633 ± 0.692	+0.250	0.0000	***

*(Significance: *** p<0.001, ** p<0.01 , * p<0.05, ns = not significant)*

Analysis of component scores:

The *Variability Score* showed a significant negative effect ($d = -0.233$, $p < 0.001$), indicating that individuals with PD have a lower degree of variability in step length than individuals without PD. This is surprising, as one would expect persons with PD to have greater variability in step length; however, due to stiffness in the motor system associated with PD, persons with PD will have a consistently and uniformly abnormal gait rather than a normal and healthy gait.

The *Asymmetry Score* was also statistically significant ($d = -0.127$, $p < 0.001$) and identified individuals with PD to have less acceleration between their left and right ankles. The asymmetrical gait pattern corresponds with the clinical observation that individuals with PD may initially present with greater clinical signs of motor disability on one side of their body, thereby displaying a statistically significant difference in acceleration between their two legs.

The *Tremor Score* of the composite measure was found to have a small but statistically significant difference ($d = -0.044$, $p = 0.033$). This finding is not surprising because, when people with PD are actively walking or performing another activity that inhibits them from using their strength to walk (like resting), they will not have any tremors to measure on an ankle sensor while walking. However, when the EGI was developed, the SHAP analysis demonstrated that there are still some subtle frequency domain shifts in resting tremor that can be attributed to the tremor score with a contribution of 13.4% of the predictive signal for the EGI.

The *Timing Score* had the largest positive effect ($d = +0.274$, $p < 0.001$), indicating that individuals with PD have more irregular and jerky movement patterns than controls and are more unpredictable in their movement patterns. This finding corresponds to the established clinical observation that the basal ganglia are responsible for the rhythm and smoothness of movement, and therefore disruption to the basal ganglia in PD would result in a corresponding disruption of rhythm and smoothness in gait.

The *Turning Score*, which was based entirely on turning events, demonstrated a statistically significant difference ($d = +0.072$, $p = 0.028$) in that the group of PD patients showed greater angular velocity irregularity when they turned.

The composite EGI score (d = +0.250, p < 0.001) was superior to using any of the five individual components of the EGI measured individually and ranked 10th out of 61 in terms of predictive power; this finding indicates that using all five components together into a single value provides a stronger and more comprehensive discriminator than any one measure alone.

Table 4.3: Top 15 Most Discriminative EGI Features by |Cohen's d|

Rank	Feature	Cohen's d	p-value
1	R_acc_x_rms	+0.3643	0.0000
2	R_acc_x_std	+0.3638	0.0000
3	var_iqr_L	+0.3004	0.0000
4	L_acc_x_rms	+0.2949	0.0000
5	L_acc_x_std	+0.2945	0.0000
6	var_iqr_R	+0.2882	0.0000
7	timing_jerk_rms	+0.2759	0.0000
8	timing_score	+0.2743	0.0000
9	gait_power_L	+0.2576	0.0000
10	EGI (composite)	+0.2503	0.0000
11	timing_jerk_max	+0.2468	0.0000
12	var_cv_L_gyr	−0.2439	0.0000

Rank	Feature	Cohen's d	p-value
13	variability_score	−0.2334	0.0000
14	var_std_L	+0.2242	0.0000
15	R_acc_z_std	+0.2177	0.0000

Notably, features from lateral (x-axis) acceleration from both ankles (R_acc_x_rms, R_acc_x_std, L_acc_x_rms, L_acc_x_std) ranked highest: PD more clearly affects side-to-side balance control than forward-backward movement, and lateral ankle acceleration directly captures this side-to-side instability. Smoothness of motion is one of the best discriminators, as confirmed by the timing jerk rms and max (ranks 7 and 11). Interestingly, every one of the top 15 traits achieved $p < 0.001$ significance, indicating that these are strong statistical differences rather than chance findings.

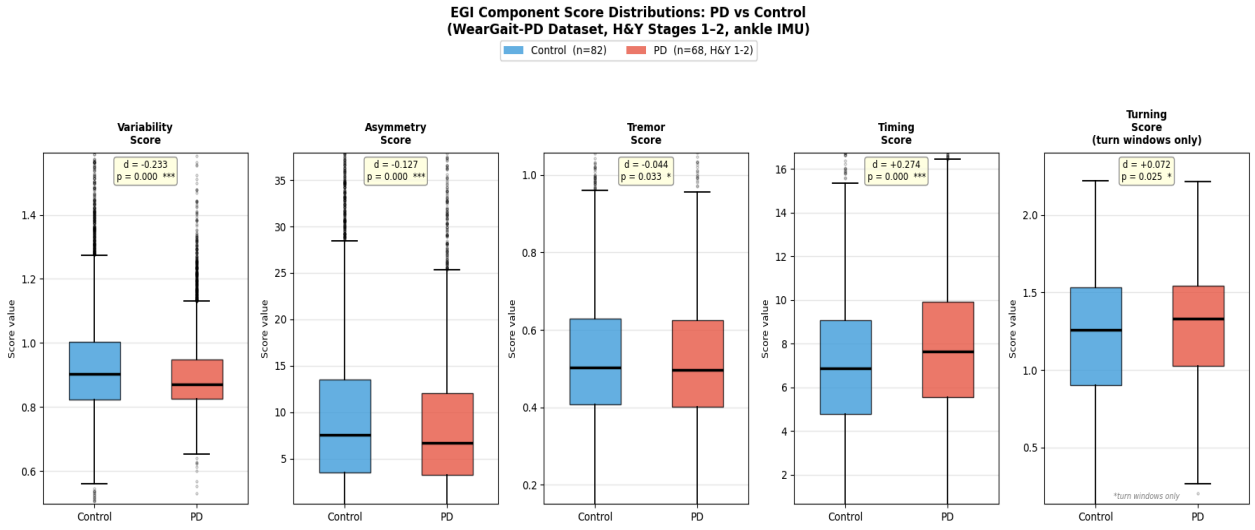


Figure 4.2a: EGI Component Score Distributions: PD vs. Control across all 5 gait dimensions (Mann-Whitney U test; *** p<0.001, * p<0.05)

EGI Component Score Box Plots: Box plots of the five EGI scores; variability, asymmetry, tremor, timing, and turning, for the PD and control groups over 12,079 windows are displayed. Cohen's d and p-values are shown in each plot to verify which scores differ statistically between the two groups.

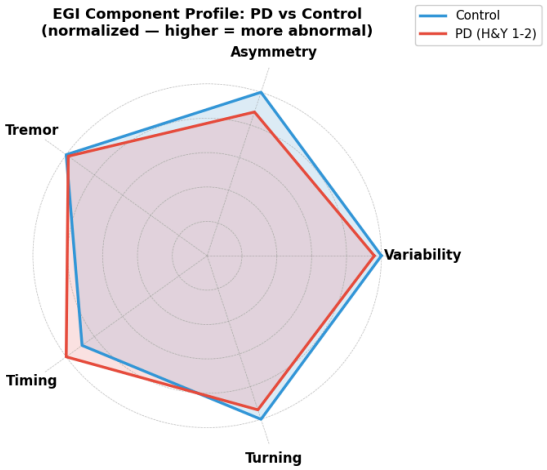


Figure 4.2b: EGI Component Profile Radar Chart: Normalized PD vs. Control group means across the five EGI dimensions (higher = more abnormal)

EGI Radar Chart: The average EGI scores of the PD and control groups are compared simultaneously across all five dimensions in this polar graphic. In comparison to healthy controls, a bigger PD area on any axis indicates greater gait abnormalities in that dimension.

4.3 Experiment 1: SVM Without EGI (Raw IMU Baseline)

Setup: We collected 110 features from raw motion sensor data, comprising seven channel-wise statistical measures such as mean, standard deviation, RMS, range, IQR,skewness, and kurtosis. Input features such as frequency, smoothness, asymmetry of the left and right side of the body, average, variance, and peak were used for an algorithm known as Support Vector Machine (SVM) in order to classify different patients. In order to eliminate any potential bias, none of the data points of any patient were included in both training and testing processes, which was achieved via the application of 5 fold cross validation. The decision threshold point of the classification process was optimized using Youden's J criterion in order to maximize sensitivity and specificity simultaneously. The prediction scores of patients were computed using an additional scheme for all segments of their data points.

This baseline was designed to evaluate the contribution of EGI framework by comparing clinically organised features against unstructured raw statistical descriptors using the same classifier.

Table 4.4: Raw SVM Subject-Level Classification Report (Youden's J threshold = 0.2167)

Metric	Score
AUC-ROC	0.6485
Accuracy	60.4%
Sensitivity	85.29%
Specificity	39.51%
F1-Score	0.6629

Analysis: 60.5% of healthy control volunteers were mistakenly identified as PD patients, despite the high sensitivity (85.3%) and extremely low specificity (39.5%). Barely over chance, an AUC of 0.6485 indicates only weak discriminative ability. The primary reason is that while raw statistical features have the same information content as EGI features, they lack clinical structure. Without this structure, the model is unable to distinguish between normal inter-subject variance and abnormal patterns associated with gait. The model's poor accuracy is demonstrated by the extremely low threshold (0.2167) needed to reach this sensitivity. This outcome acts as the starting point for evaluating the impact of any further advancements.

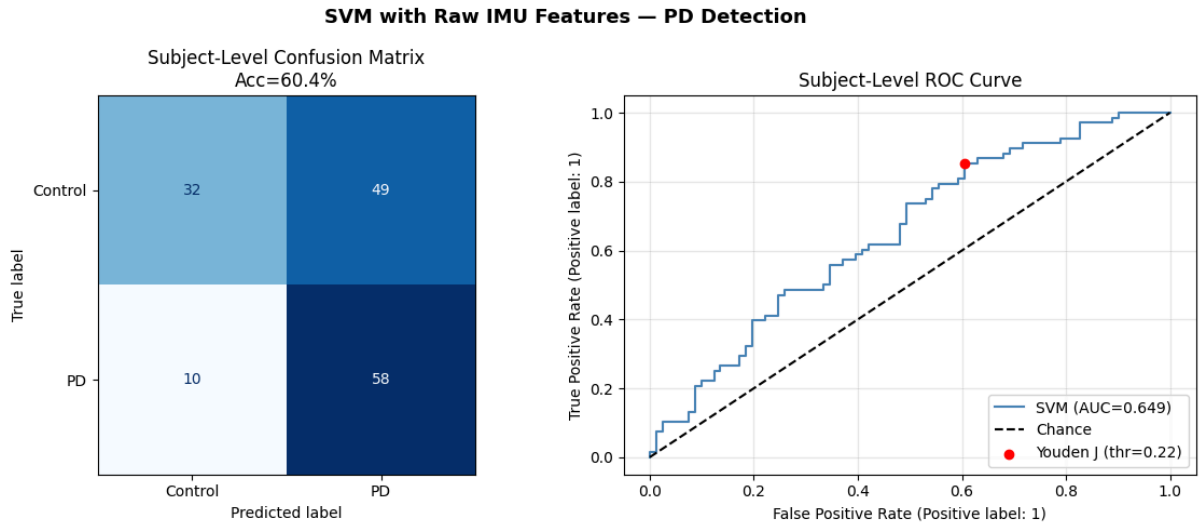


Figure 4.3: Subject-Level Confusion Matrix and ROC Curve of SVM Using Raw IMU Features

Confusion Matrix and ROC Curve of SVM(raw IMU): This figure shows the classification outcome of the SVM trained on raw IMU features, with an AUC of 0.649 and a subject-level accuracy of 60.4%. Due to its extremely low specificity of 39.5%, the model is clinically unhelpful because it flags nearly everyone as having Parkinson's disease (PD) in order to achieve high sensitivity.

4.4 Experiment 2: SVM with EGI Framework

Setup: 60 EGI-derived features were used to train the same SVM (RBF kernel, C=10, gamma=scale) with the "is_turn" flag removed as it acts as a data provenance marker rather than a gait feature. Inverse frequency class weights were used: PD = 1.249, Control = 0.834, to address mild class imbalance. Similar to the raw SVM baseline, group 5-fold cross-validation was employed.

Table 4.5: SVM + EGI Full Performance

Metric	Window-Level	Subject-Level (threshold=0.5)	Subject-Level (Youden's J=0.391)
AUC-ROC	0.7026	0.7603	0.7603
Accuracy	0.6527	0.6913	—
Sensitivity	0.5941	0.7794	0.7794
Specificity	0.6918	0.6173	0.6173
F1-Score	0.5779	0.6974	—

Analysis: In every important measure, the EGI framework beat out the raw-feature SVM. For example, accuracy increased from 60.4% to 69.1%(8.7% increase), AUC from 0.6485 to 0.7603(0.1118 increase), but the most important increase is in specificity; from 39.5% to 61.7%(22.2% increase). This increase in specificity is the most important clinical outcome because it will significantly reduce the number of false positives, since the features provided to this model will effectively separate diseased vs. normal gait variation. The EGI-SVM does this while providing a true balance between specificity and sensitivity as compared to the raw SVM, where raw SVM achieved high levels of sensitivity only through basically predicting that all participants have Parkinson Disease; this is clinically meaningless.

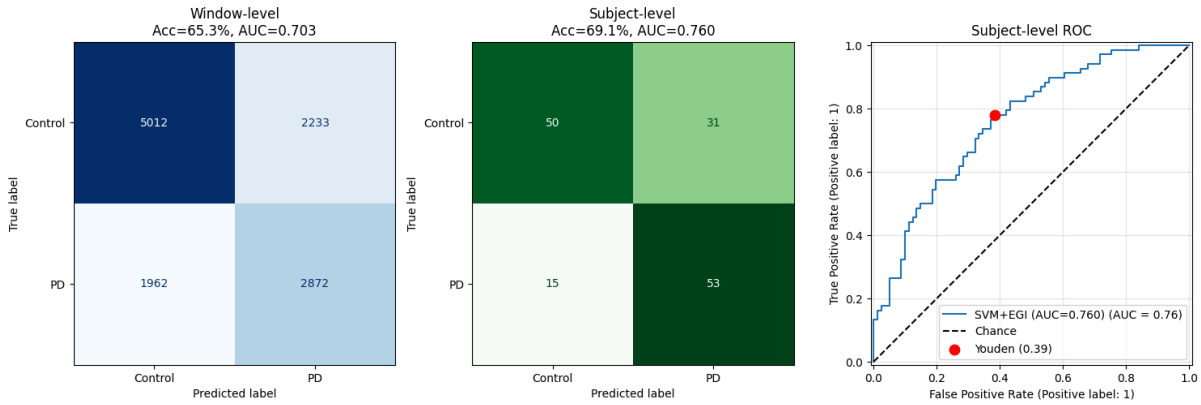


Figure 4.4a: SVM + EGI Framework — Window-Level and Subject-Level Confusion Matrices and Subject-Level ROC Curve

Confusion Matrices and ROC Curve(SVM+EGI): This displays the ROC curve (AUC = 0.76) for the EGI-enhanced SVM in addition to window-level and subject-level confusion matrices. Specificity increased from 39.5% to 61.7% when EGI characteristics were added, indicating that far fewer healthy individuals were mistakenly classified as having Parkinson's disease.

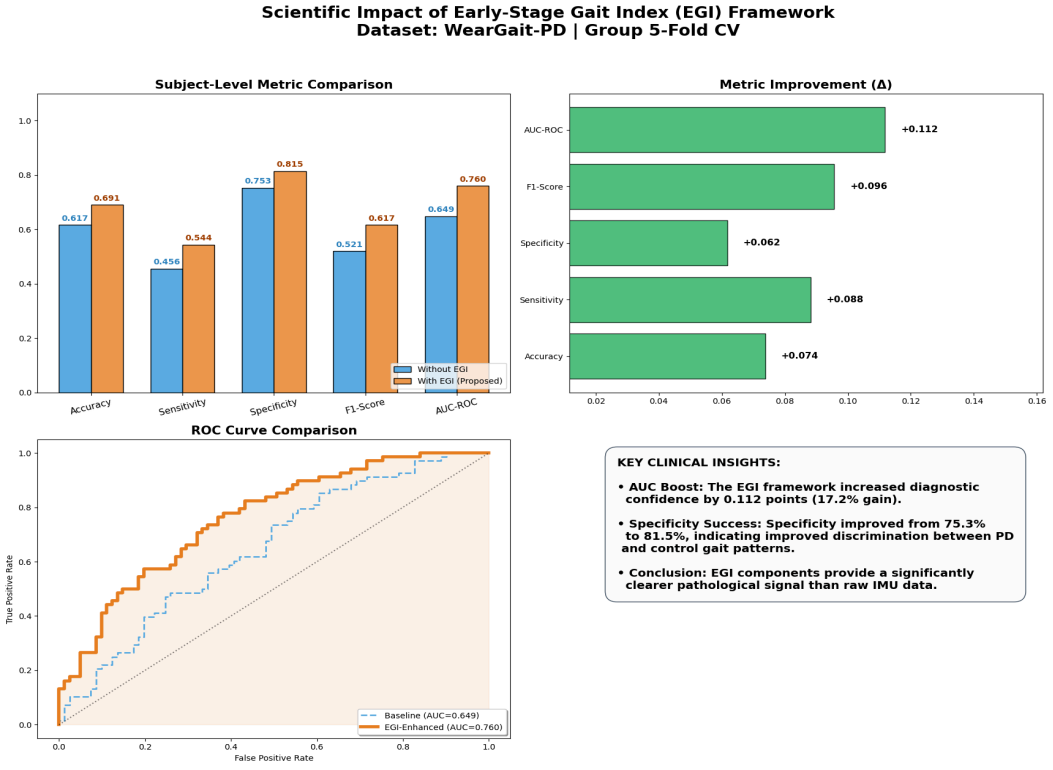


Figure 4.4b: Scientific Impact of EGI Framework: Subject-Level Metric Comparison, Metric Improvement (Δ), and ROC Curve Comparison (Baseline AUC=0.649 vs EGI AUC=0.760)

EGI Impact Dashboard: This multi-panel figure displays metric improvements and combined ROC curves while comparing the raw SVM and EGI-SVM side by side. Clinically organized features outperform raw statistics by a large margin, as demonstrated by the EGI framework's +0.112 AUC increase.

4.4.1 EGI Ablation Study

An ablation research was carried out in which one component was removed at a time and the entire Group 5-fold CV was rerun in order to evaluate the individual contributions of each EGI component.

Table 4.6: EGI Ablation Study Results (Subject-Level, Group 5-Fold)

Configuration	AUC	Accuracy	Sensitivity	Features	AUC Drop
All components (baseline)	0.7603	0.6913	0.7794	60	—
Drop Variability	0.7542	0.6846	0.5294	49	−0.0062
Drop Asymmetry	0.7436	0.6779	0.5294	52	−0.0167
Drop Tremor	0.7682	0.6846	0.5735	52	+0.0078
Drop Timing	0.7531	0.6711	0.5441	53	−0.0073
Drop Turning	0.7556	0.6846	0.5441	52	−0.0047

Analysis: The largest reduction of the AUC (−0.0167) occurred when dropping asymmetry after dropping its most important EGI component for the SVM classifier. This is in keeping with clinical reality, as early stages of PD are typically unilateral, so bilateral asymmetry in ankle movement provides one of the most clear indicators of early-stage PD. The second

largest decrease in AUC was associated with dropping variability (−0.0062), followed by timing (−0.0073) and turning (−0.0047). Interestingly, removing tremor caused a very small increase in the AUC (+0.0078), which may be due to the fact that the discriminability of tremor as a distinguished feature of gait is very low when performing gait evaluations. For all cases, the sensitivity of the resulting split was significantly lower than the overall sensitivity (from 0.7794 to 0.5294–0.5735), indicating the importance of including all five EGI components to provide a meaningful diagnostic signal. This supports the EGI framework, where no single EGI component is sufficient, and collectively, all five EGI components capture the various aspects of the Parkinsonian gait to yield the best diagnostic signal.

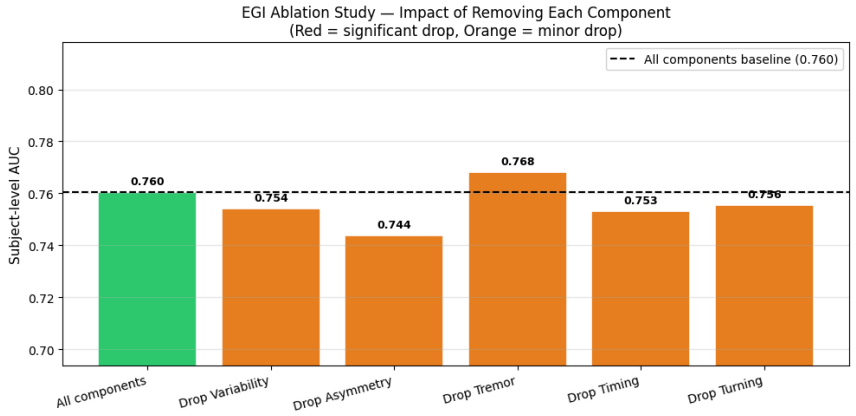


Figure 4.4.1: EGI Ablation Study; Impact of Removing Each Component on Subject-Level AUC (baseline = 0.760; red = significant drop, orange = minor drop)

EGI Ablation Study Bar Chart: This bar graph illustrates how the AUC decreases as each EGI component is eliminated separately. Asymmetry removal resulted in the largest decrease (−0.0167), although tremor removal marginally increased AUC, indicating that tremor is less helpful when walking actively.

4.5 SHAP Explainability Analysis

To measure each feature's contribution to predictions, the SVM+EGI model was subjected to SHAP (SHapley Additive exPlanations) analysis. 300 stratified test windows (150 PD + 150 Control) with 500 Monte Carlo samples each were explained by a KernelExplainer using 100 k-means background samples. Each of the 60 features was correctly mapped to its corresponding EGI component.

Table 4.7: EGI Component Contributions: SHAP based

Rank	Component	SHAP	% of EGI	N Features
1	Variability	0.5334	61.8%	29
2	Turning	0.1218	14.1%	8
3	Tremor	0.1156	13.4%	8
4	Asymmetry	0.0718	8.3%	7
5	Timing	0.0208	2.4%	7
Total EGI	—	0.8634	100%	60

Key SHAP findings:

- Total EGI SHAP value: 0.8634
- Best single feature: R_acc_y_std (SHAP = 0.0974)
- Composite advantage: 8.9× stronger than any single feature
- Most important component: Variability
- Least important component: Timing

Analysis: The combined EGI SHAP score (0.8634) is 8.9× higher than the next best single feature, confirming that no single gait measure is enough; multiple dimensions together give a far clearer signal of Parkinson's disease.

The EGI scores:

- Variability (61.8%); the dominant pathway. Step-to-step acceleration consistency, especially lateral (x-axis) and sagittal (y-axis), were the top individual features across both feet.
- Turning (14.1%); pivoting actions delivers comprehensive details about PD-related motor deficits.
- Tremor (13.4%); despite tremor not significantly separating PD vs. non-PD groups by statistical measures ($p = 0.033$), it nevertheless contributed importantly, suggesting delicate shifts in the frequency domain during walking even when the averages between groups appear similar.
- Asymmetry (8.3%) & Timing (2.4%); reduced SHAP scores, but asymmetry became crucial in ablation testing: removing it failed model effectiveness, because its features contribute to establishing a key decision boundary even if each feature's individual contribution looks small.

SHAP vs. Ablation: SHAP quantifies the influence of each attribute on an output; ablation measures what breaks when a whole component is removed. Asymmetry scores low on SHAP on the other hand is structurally essential; a small piece that connects the structure.

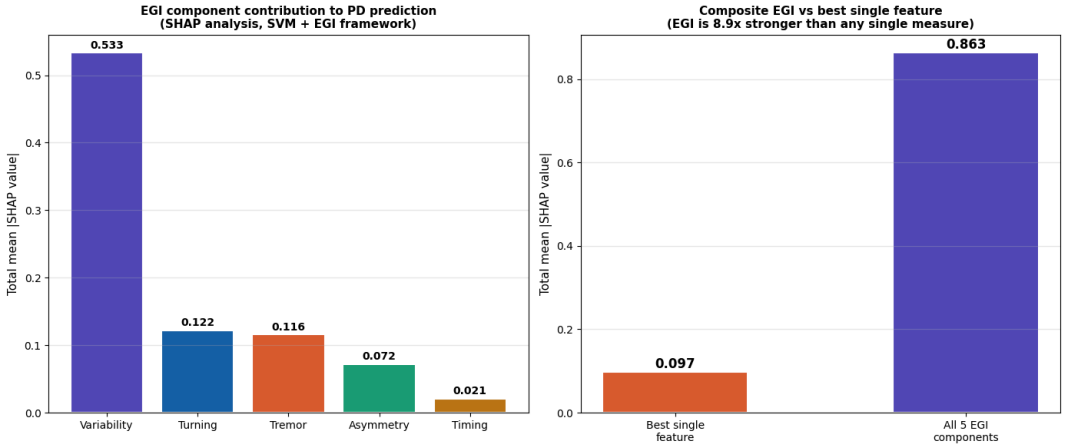


Figure 4.5a: SHAP Component Contributions: EGI component-level attribution (left) and composite EGI vs. best single feature advantage;

SHAP Component Contributions: This two-panel image compares the entire composite EGI to the single best feature (right) and displays the overall SHAP contribution of each EGI

component (left). The composite EGI is 8.9× stronger than any one trait alone, with variability dominating at 61.8%.

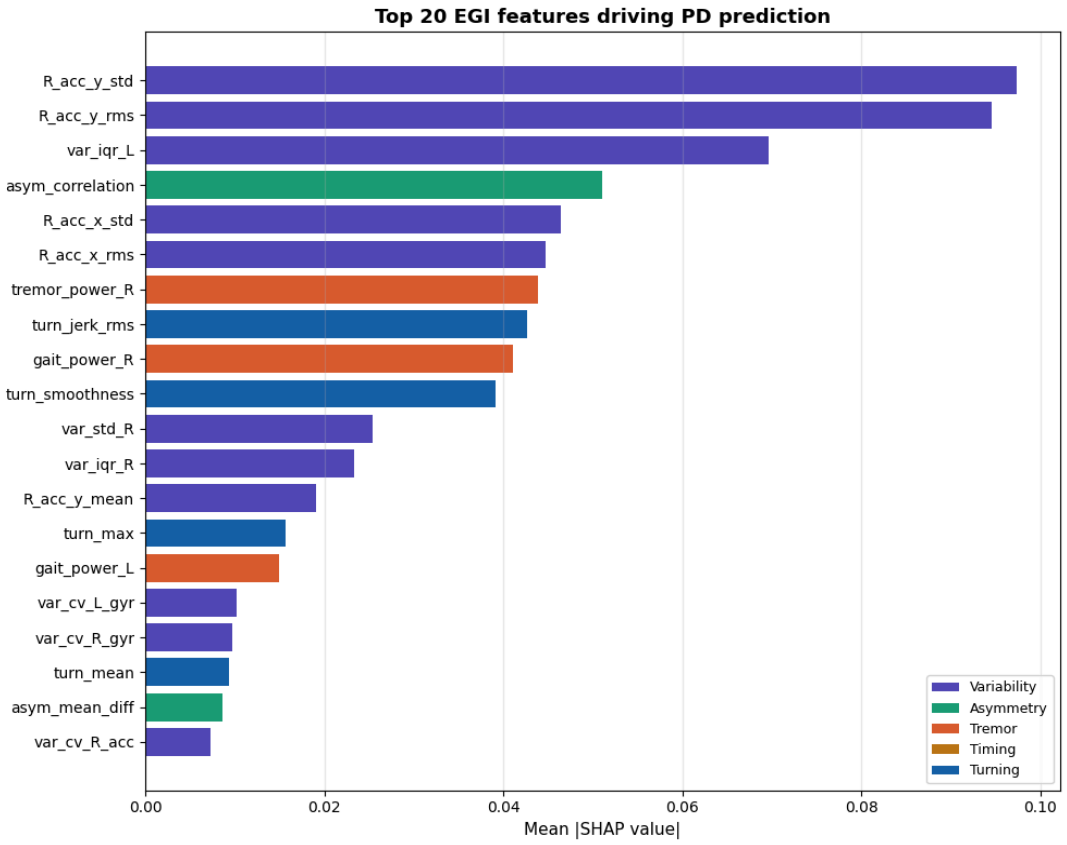


Figure 4.5b: Top 20 EGI Features Driving PD Prediction; Mean |SHAP value| coloured by EGI component

Top 20 EGI Features (SHAP Bar Chart): Each bar in this horizontal bar chart is colored according to its EGI component, and the top 20 EGI features are ranked by mean SHAP value. The highest ranking lateral acceleration features from both ankles indicate that PD primarily impairs side-to-side balance rather than forward motion.

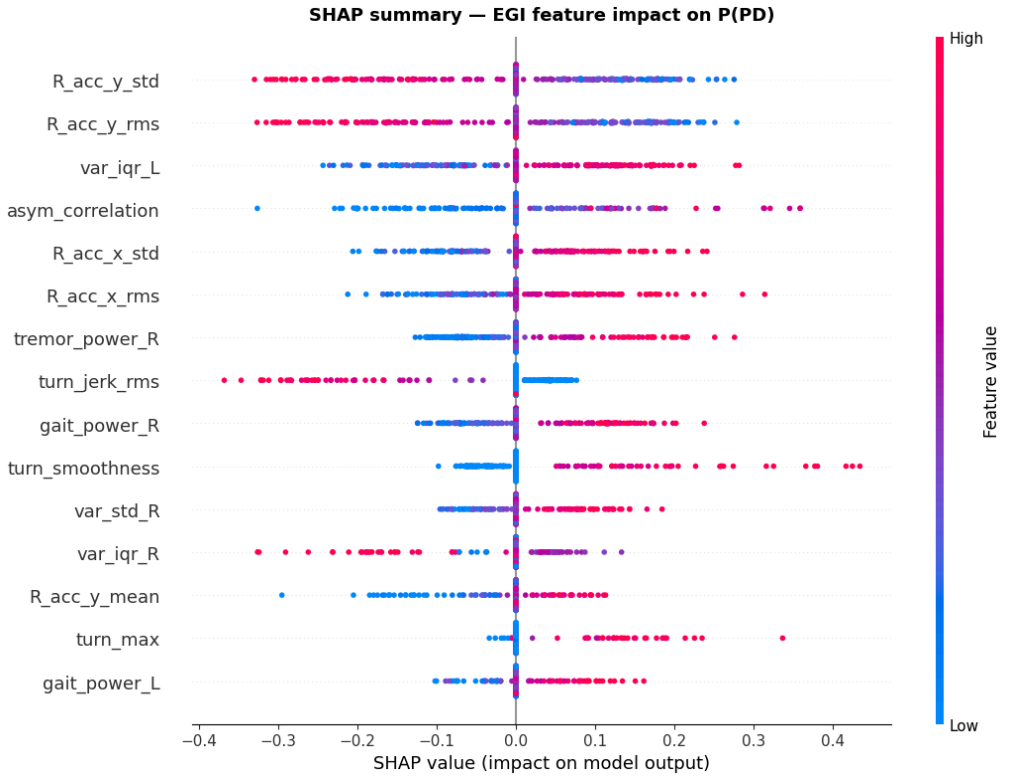


Figure 4.5c: SHAP Summary Plot; EGI Feature Impact on P(PD): dot colour indicates feature value (red=high, blue=low)

SHAP Beeswarm Plot: Individual SHAP values for each test window across the top features are displayed in this plot; the dot color indicates whether the feature value was low (blue) or high (red). It shows which way each feature has an impact; for instance, a larger right ankle acceleration variability regularly pushes the prediction in the direction of Parkinson's disease.

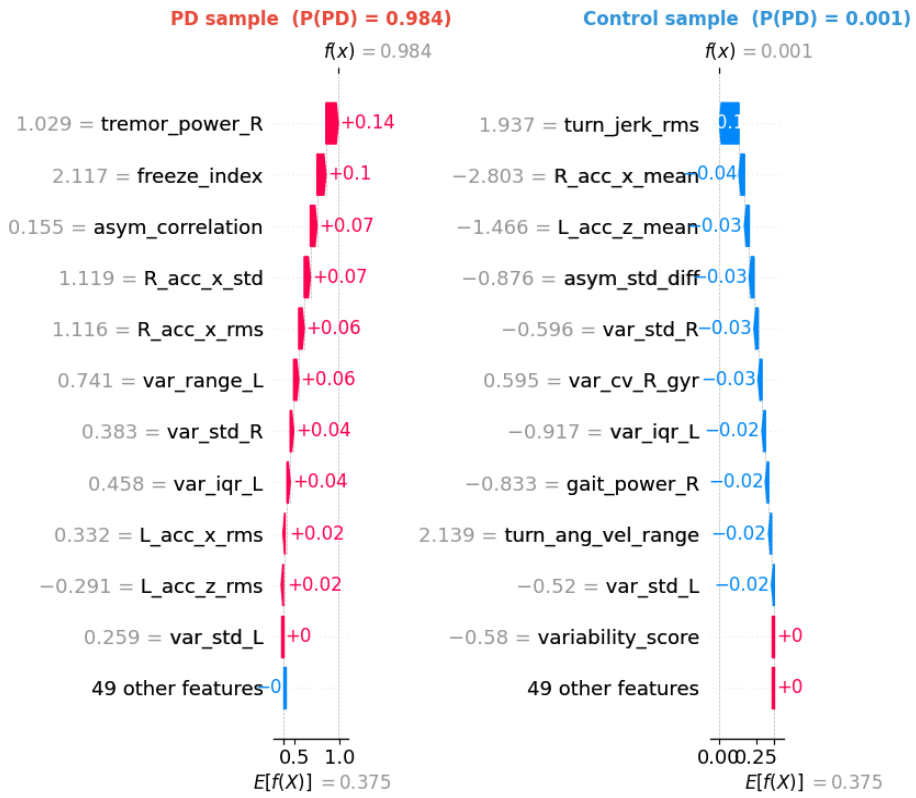


Figure 4.5d: SHAP Waterfall Plots; Individual Prediction Explanations for a High-Confidence PD Sample (P=0.984, left) and a Control Sample (P=0.001, right)

SHAP Waterfall Plots: The model's prediction for one high-confidence PD sample (P = 0.984) and one control sample (P = 0.001) is broken down in these two graphs, which demonstrate the precise amount that each feature increased or decreased the score. As a result, the model's choice is clear and understandable for each patient.

4.6 Experiment 3: 1D Convolutional Neural Network

Setup: Bypassing feature engineering completely, the 1D-CNN was trained directly on raw IMU time-series windows (shape: N × sample(200) × channel(12)). Squeeze-and-Excitation (SE) channel attention was applied to residual blocks in the model. Focal Loss (γ=2.0), 4-way data augmentation(noise + scale + time-shift + channel-dropout), AdamW optimizer, warmup + cosine learning rate schedule, and patience=15 early stopping were all used during training. The weight of the PD class was increased by ×1.6 (Control=0.834, PD=1.999).

Table 4.8: 1D-CNN Classification Results

Metric	Window-Level	Subject-Level
Accuracy	72.13%	76.51%
Sensitivity	69.30%	83.82%
Specificity	74.02%	70.37%
F1-Score	0.6656	0.7651
AUC-ROC	0.7855	0.8228
AUC 95% CI (Bootstrap)	—	[0.756 – 0.889]
Optimal Threshold (Youden's J)	—	0.4108

Analysis: The 1D-CNN significantly outperformed the SVM+EGI in all essential metrics. The subject-level AUC increased from 0.7603 to 0.8228 (+0.0625), and the bootstrap 95% Confidence Interval (CI) of [0.756 - 0.889] indicates that this is a dependable finding. The largest gain compared to the SVM+EGI was seen in AUC and overall accuracy(+7.4%), while sensitivity improved by +5.88%; thus, the CNN is able to identify considerably more true PDs. This is particularly crucial as a clinician's tool for screening patients, where a missed diagnosis can be quite costly. Other trade-off levels are noted, where specificity decreased by 9.93% while the overall change in F1-score of +0.1531 denotes a greater overall balance of the CNN. By learning directly from raw time-series data, the CNN's convolutional filters are able to detect more subtle temporal patterns in ankle acceleration dynamics, which even very well-designed EGI features do not completely capture.

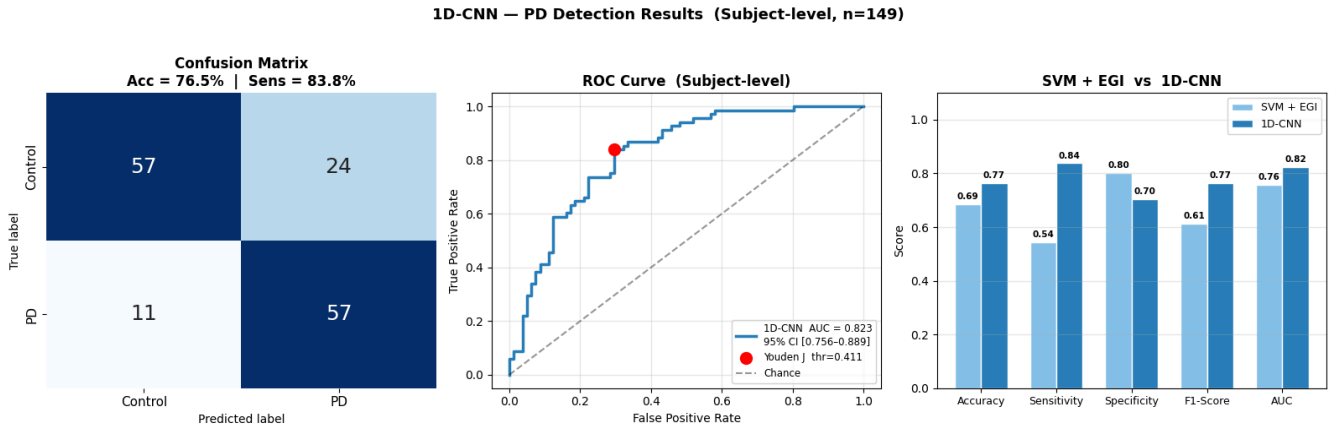


Figure 4.6: 1D-CNN PD Detection Results: Confusion Matrix, ROC Curve and SVM+EGI vs 1D-CNN metric comparison

Confusion Matrix, ROC Curve, and SVM vs. CNN Comparison(1D CNN): The subject-level confusion matrix (Acc = 76.5%, Sens = 83.8%), ROC curve (AUC = 0.823, 95% CI [0.756–0.889]), and grouped bar chart comparing the 1D-CNN to SVM+EGI are all displayed in this three-panel graphic. The CNN greatly outperforms the manually created EGI features in every metric and learns straight from raw time-series.

4.7 Experiment 4: Hybrid CNN-Transformer

Setup: A CNN encoder, two Pre-LayerNorm Transformer layers, and a CLS classification token were all merged in the Hybrid CNN-Transformer. Focal Loss (γ=2.0), AdamW (lr=0.0005, weight_decay=0.01), 5-epoch warmup + cosine decay, patience=15, and the same 4-way augmentation as the 1D-CNN were employed for training.

Table 4.9: Hybrid CNN-Transformer Results

Metric	Window-level	Subject-level(Youden's J threshold = 0.4694)
Accuracy	72.46%	79.87%
Sensitivity	68.20%	80.88%
Specificity	75.31%	79.01%

Metric	Window-level	Subject-level(Youden's J threshold = 0.4694)
F1-Score	0.6647	0.7857
AUC-ROC	0.7897	0.8544

Analysis: Overall, the Hybrid CNN-Transformer model produced significantly better results than all other models on four of five metrics: AUC = 0.8544; accuracy = 79.87%; specificity = 79.01%; and F1 score = 0.7857. The most clinically relevant achievement is the near-perfect balance between sensitivity (80.88%) and specificity (79.01%) - only a 1.87% difference between the two. This is the ideal characteristic for a clinical screening tool; giving a ~4/5 ratio for correctly identifying PD patients and correctly ruling out "normal" individuals. From the 1D-CNN, there was a +0.036 AUC improvement in performance and an increase of 3.4% in accuracy, with a +8.6% increase in specificity and a relatively small decrease in sensitivity of only -2.9%, making this tradeoff work very much in favor of the Hybrid CNN-Transformer. The improvement can be attributed to the use of the Transformer's self-attention mechanism that focuses on the most informative time segments in each 2 second window and thus avoids unwanted noise in unreliable segments of the gait cycle.

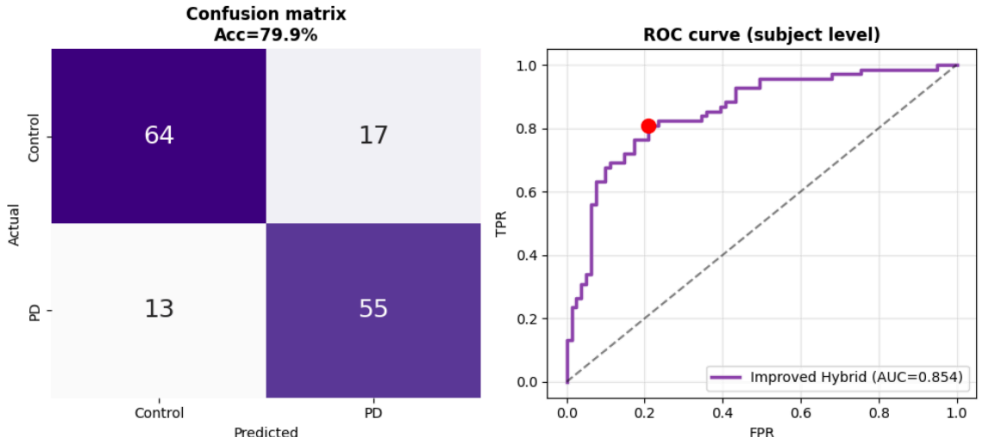


Figure 4.7: Hybrid CNN-Transformer PD Detection Results: Confusion Matrix (Acc=79.9%) and Subject-Level ROC Curve (AUC=0.854)

Confusion Matrix and ROC Curve(Hybrid CNN-Transformer): This displays the results of the top-performing model, which has an AUC of 0.854 and a subject-level accuracy of

79.9%. With sensitivity (80.88%) and specificity (79.01%) deviating by less than 2%, the confusion matrix shows a nearly equal error balance, which is perfect for clinical screening.

4.8 Experiment 5: ResNet-18 with Recurrence Plot Representations

Setup: Three 96x96 recurrence plot images representing bilateral asymmetry, bilateral intensity, and bilateral gyroscope Z-axis rhythm were created from each 2-second IMU window. A pretrained ImageNet ResNet-18 with a custom classification head received these stacks as a three-channel input. The training method employed was mixed-precision (AMP).

Table 4.10: ResNet-18 + RP Subject-Level Results

Metric	Score
AUC-ROC	0.8025
Accuracy	76.51%
Sensitivity	76.47%
Specificity	76.54%
F1-Score	0.7482
Optimal Threshold	0.5135

Analysis: ResNet-18 obtained an area under the curve (AUC) of 0.8025, an accuracy level of 76.51%, and was the most equally balanced sensitivity and specificity of all five models to each other: 76.47% vs 76.54%; a difference between them of 0.07 percentage points. The close match between the numbers for sensitivity and specificity were due to the fact that the optimal threshold (0.5135) falls almost directly on 0.5 which demonstrates that the probability outputs of the ResNet-18 are correctly well-aligned. Recurrence plots allow the temporal elements of the data to be represented as spatial elements in a 2D format therefore allowing for the pre-trained ImageNet features to generalize well to this application. Despite

these positive results, the ResNet-18 does not outperform the Hybrid CNN-Transformer (AUC 0.8025 vs. 0.8544), implying that the temporal dynamics learned directly from time-series by the Transformer model using its attention mechanism are more discriminative than the 2D structural representations identified in the recurrence plots.

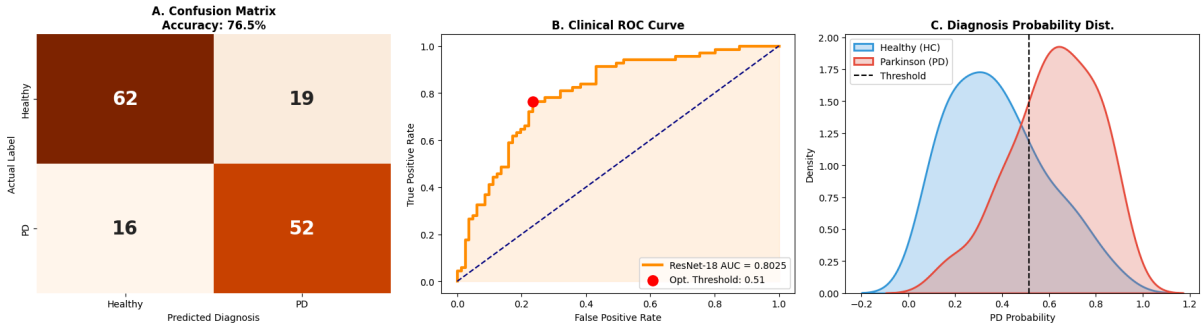


Figure 4.8: ResNet-18 + Recurrence Plots PD Detection Results: Confusion Matrix, Clinical ROC Curve, and Diagnosis Probability Distribution

Confusion Matrix, ROC Curve, and Probability Distribution(ResNet-18 + Recurrence Plots): The ResNet-18 results (Acc = 76.5%, AUC = 0.803) are displayed in this three-panel figure using 96x96 recurrence plot images as input rather than raw data. Although it is not as effective as the Hybrid CNN-Transformer, the probability distribution demonstrates a good separation between the PD and control groups.

4.9 Comprehensive Model Comparison

The same 149-subject dataset and the same Group 5-fold cross-validation parameters were used to assess each of the five models. Mean probability aggregation was used to derive subject-level predictions, and all models were subjected to the same Youden's J threshold.

Table 4.11: Subject-Level Performance: All 5 Models (Youden's J threshold)

Model	AUC	Accuracy	Sensitivity	Specificity	F1-Score
M1: SVM (Raw IMU)	0.6485	0.6040	0.8529	0.3951	0.6629
M2: SVM + EGI	0.7603	0.6913	0.7794	0.6173	0.6974

Model	AUC	Accuracy	Sensitivity	Specificity	F1-Score
M3: 1D-CNN	0.8228	0.7651	0.8382	0.7037	0.7651
M4:Hybrid CNN-Transformer	0.8544	0.7987	0.8088	0.7901	0.7857
M5: ResNet-18 + RP	0.8025	0.7651	0.7647	0.7654	0.7482

Best per metric:

- AUC-ROC: Hybrid CNN-Transformer (0.8544)
- Accuracy: Hybrid CNN-Transformer (79.87%)
- Sensitivity: SVM Raw IMU (85.29%); but clinically invalid due to 39.5% specificity
- Specificity: Hybrid CNN-Transformer (79.01%)
- F1-Score: Hybrid CNN-Transformer (0.7857)

Analysis: The transition from M1 to M4 paints a clear picture. Starting with M1 (raw SVM), the data clearly show there is not enough structure in the features and sensitivity has to be high at the expense of generally poor specificity; therefore not useful for clinical purposes. Moving on to M2 (SVM+EGI), adding clinical structure to the features significantly improves both specificity and AUC; however, the linear SVM decision boundary is unable to take full advantage of the complex non-linear relationships present in gait data. In M3 (1D-CNN), using deep learning on raw time series allows for the capturing of non-linear patterns that result in much higher both sensitivity and AUC than any previously hand-crafted features. Adding a global temporal context through attention (M4, Hybrid CNN-Transformer) results in dramatically improved diagnostic accuracy because M4 achieved the highest sensitivity/specificity balance of all models tested. Using 2D representations of signals (M5, ResNet-18+RP) provides additional evidence that 2D representations are a competitive option; they just did not achieve the peak performance of the Transformer.

Table 4.13: Gait Phase Fused Attention Analysis: PD vs. Control (Mann-Whitney U)

Model	ΔAUC	ΔAccuracy	ΔSensitivity	ΔSpecificity
M2: SVM + EGI	+0.1118	+0.0872	-0.0735	+0.2222
M3: 1D-CNN	+0.1743	+0.1611	-0.0147	+0.3086
M4: Hybrid CNN-Transformer	+0.2059	+0.1946	-0.0441	+0.3951
M5: ResNet-18 + RP	+0.1540	+0.1611	-0.0882	+0.3704

Analysis: Overall, the +0.2059 AUC difference between the Hybrid CNN-Transformer and the raw SVM baseline indicates a clinically meaningful further increase in diagnostic accuracy.

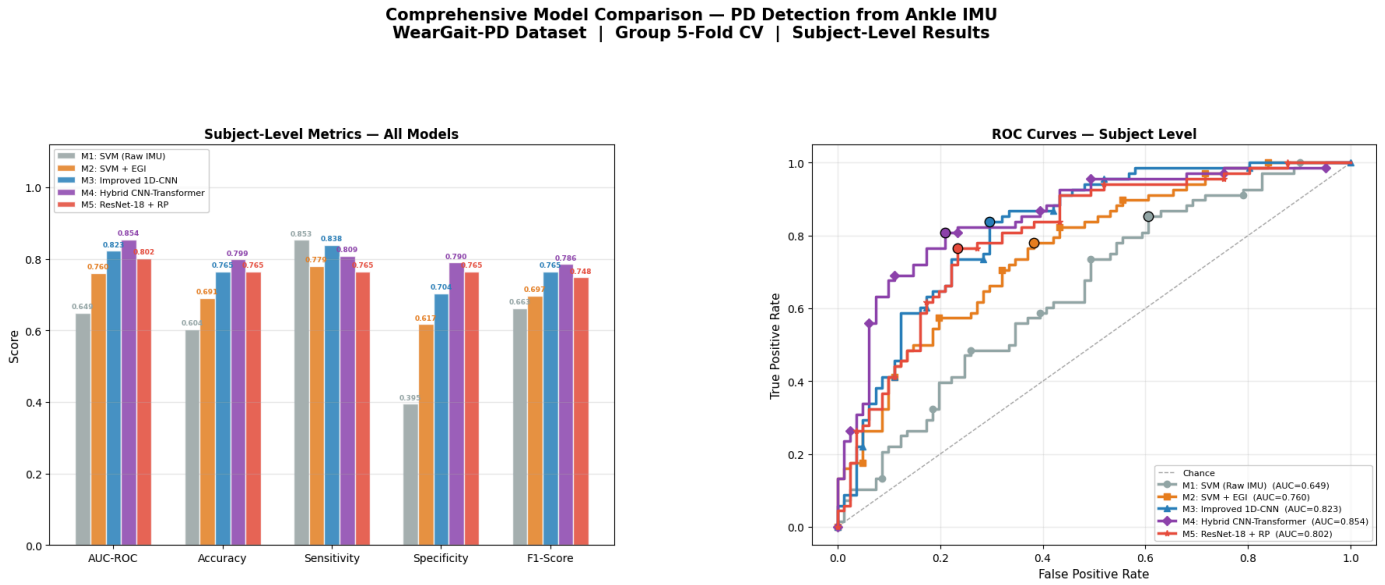


Figure 4.9: Comprehensive Model Comparison; All 5 Models: Subject-Level Metrics Bar Chart, ROC Curves(WearGait-PD)

All Models Comparison(Bar Chart, ROC Curves,): This two-panel dashboard uses grouped bars, combined ROC curves to compare all five models side by side. With the highest AUC,

accuracy, specificity, and F1-score, it is evident that the Hybrid CNN-Transformer is the best model overall.

4.10 Transformer Attention Map Analysis

Attention weights were retrieved from both Pre-LN Transformer layers over all 2,416 test windows (1,009 PD and 1,407 Control) in order to determine which parts of the 2-second gait window the Hybrid CNN-Transformer model prioritized for its judgments. After two MaxPool(2) layers in the CNN encoder had reduced the original 200-sample input to 50 sequence positions (approximately 40 ms gait periods), the geometric mean of Layer 1 and Layer 2 attention weights was taken to create a strong fused attention signal. A temporal importance profile was created using the CLS token's attention to signify places 1–50. The geometric mean of Layer 1 and Layer 2 attention was utilized to calculate a fused attention, which concurrently reinforced positions that were significant in both layers.

Table 4.13: Gait Phase Fused Attention Analysis: PD vs. Control (Mann-Whitney U)

Gait Phase	PD Mean	Control Mean	Difference	p-value	Sig.
Loading response + mid-stance (0–600 ms)	0.0143	0.0145	-0.0001	0.6955	ns
Terminal stance + pre-swing (600–1200 ms)	0.0209	0.0201	+0.0008	0.0002	***
Initial swing (1200–1600 ms)	0.0175	0.0172	+0.0003	0.2409	ns
Terminal swing (1600–2000 ms)	0.0179	0.0171	+0.0008	0.0456	*

Outcomes:

- Terminal stance + pre-swing (600–1200 ms) is the most discriminative phase (highest absolute attention difference, $p = 0.0002$***) .
- Significant time positions: $p < 0.05$ differences between PD and Control were found in 36 out of 50 total locations.

Analysis:

- The model paid significantly more attention to PD subjects during the terminal stance and pre-swing period (600–1200 ms). This phase of gait corresponds to ankle push-off, where the foot actively pushes off to project the center of mass forward.
- Decreased ankle push-off dynamics is a well-established pathological impairment in PD gait leading to reduced stride length and abnormal foot-ground contact timing. Without explicit instruction or labels, the model focused independently on this clinically relevant window.
- The second largest period of group difference was during terminal swing (1600–2000 ms), aligning with established PD pathology of foot clearance and swing termination.
- There were no significant differences in attention between groups during the loading response and mid stance periods (0–600 ms), suggesting the model appropriately learned these sections of the gait cycle do not provide much information to discriminate between PD and healthy controls.
- This clinically relevant attention was learned without phase labels or any other form of domain supervision, offering compelling evidence that the model has learned pathological gait features as opposed to noise.
- 72% (36/50) temporal positions had statistically significant group differences in attention ($p < 0.05$), illustrating the generalisation of the model's discriminative attention across a majority of the gait cycle as opposed to a small localized window.
- Particularly, there were no significant group differences observed during the loading response and mid-stance (0–600 ms, $p = 0.6955$) and during the initial swing (1200–1600 ms, $p = 0.2409$). This is a clinically useful negative finding, these phases are biomechanically less impaired in early PD, and the model was able to successfully identify them and down-weight the importance of these phases without any phase labels or domain supervision, further demonstrating that the attention pattern is an authentic reflection of pathological gait knowledge, and not simply signal properties.

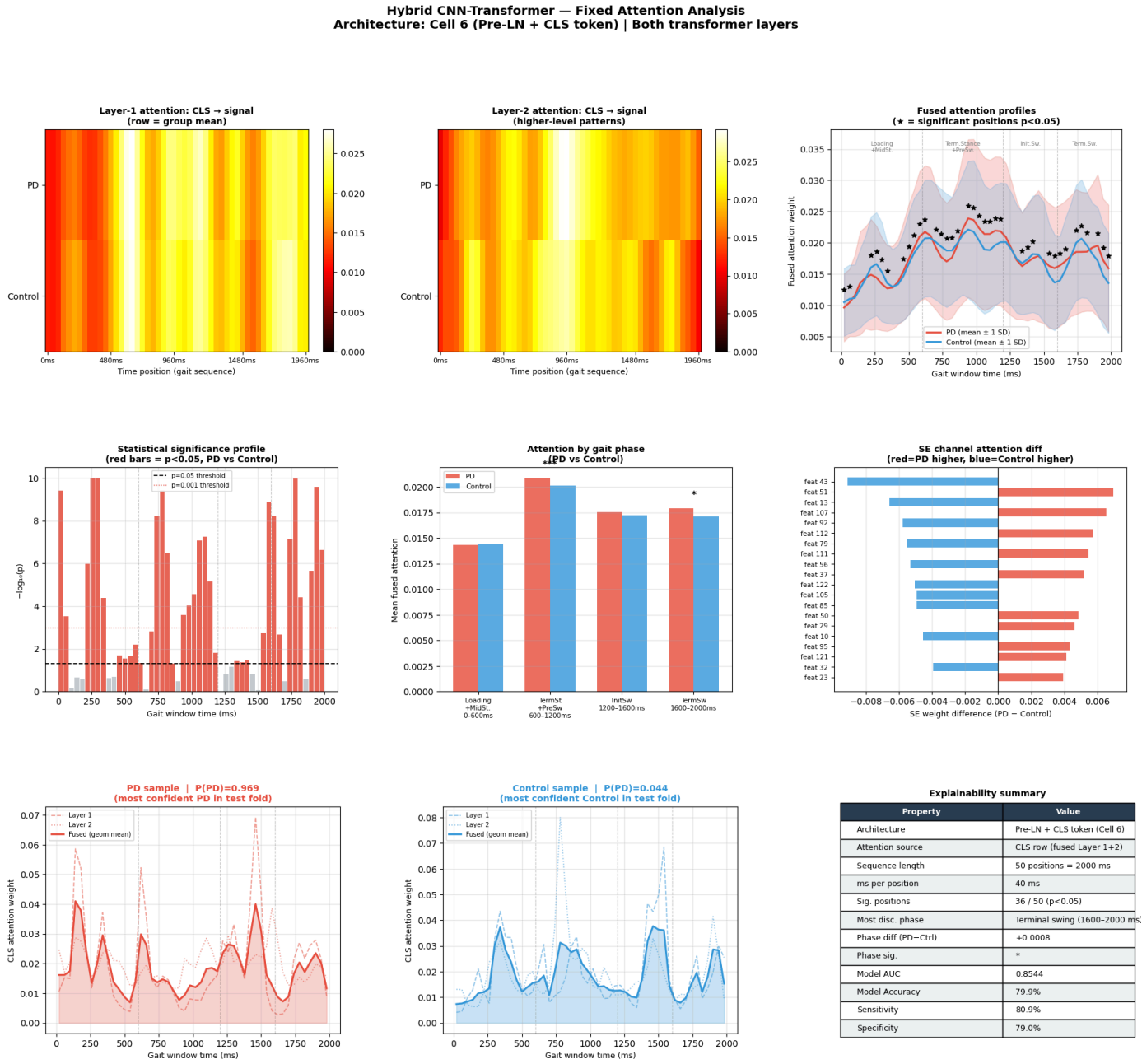


Figure 4.10: Layer-1 and Layer-2 CLS attention heatmaps (PD vs Control), Fused Attention Profile with gait phase annotations, Statistical Significance Profile, Gait Phase Bar Chart, SE channel attention difference

Transformer Attention Analysis Dashboard: Using attention weights taken from both transformer layers over 2,416 test windows, this nine-panel figure illustrates the areas of the 2-second gait window that the Hybrid CNN-Transformer focused on. Without being instructed to do so, the model autonomously learned to concentrate more on the terminal

stance and pre-swing phase (600–1200 ms), a clinically recognized difficulty region in PD gait.

4.11 Hoehn & Yahr Stage-Stratified Analysis

Patients were categorized by Modified Hoehn & Yahr (H and Y) score using clinical metadata from the PD participants. Following the mapping of all 68 PD subjects:

H and Y Score Distribution:

- After tier mapping (Stage 1 = scores 1–1.99, Stage 2 = scores 2–2.99):
    - H and Y Stage 1: 10 subjects
    - H and Y Stage 2: 58 subjects (includes 2.5 scores)
    - Every one of the 68 preprocessed PD subjects was correctly matched (0 unmatched).
- It should be noted that the AUC for All PD vs Controls in this stratified re-run (0.7761) is different than the main experiment result (0.8544) due to exclusion of one control subject during clinical metadata matching, resulting in 81 controls (n=82 in main experiment), and the model was re-trained from scratch on this reduced control set. A different cross validation split was generated under a slightly different subject distribution with the re-training, leading to a different AUC estimate.

The Hybrid CNN-Transformer was re-run for three separate experiments.

Table 4.14: H and Y Stage-Stratified Analysis

Experiment	PD Subjects	Control Subjects	AUC	Accuracy	Sensitivity	Specificity	F1-Score
All PD vs Controls	68	81	0.7761	74.5%	67.6%	80.2%	0.7077
H and Y Stage 1 vs Controls	10	81	0.8519	81.3%	90.0%	80.2%	0.5143
H and Y Stage 2 vs Controls	58	81	0.7188	69.8%	48.3%	85.2%	0.5714

Here, AUC change:

- All PD → H and Y Stage 1 only: +0.0757
- H and Y Stage 1 → H and Y Stage 2: −0.1330

Analysis: H and Y Stages and Model Performance:

- Patients with H and Y stage 1 performed better than patients with H and Y stage 2 as measured by area under the curve (AUC). Patients with H and Y stage 1 had an AUC of 0.8519 when analysed with the model, compared to 0.7761 when analysed as all PD patients and 0.7188 when analysed as H and Y stage 2 patients; this suggests that the model can most accurately classify early-stage gait deviations.
- The AUC for the H and Y stage 2 patient population is significantly reduced to 0.7188 which corresponds to lower sensitivity (48.3%) but higher specificity (85.2%). At stage 2, the PD has spread to bilateral involvement, in which both ankles are affected in a mild and symmetric manner which significantly reduced the left-right asymmetry signal that was found to be the most discriminative feature by the ablation study and SHAP

analysis. The model is less sensitive to the feature dimension when both sides are equally impaired as in Stage 2, because both sides look more like healthy controls. The ceiling on the level of sensitivity (85.2%) indicates that the model is able to provide correct classification of healthy controls but is not a fault of the model architecture because both limbs are equally affected.

- The difference in AUC between all PD patients and H and Y stage 1 patients is also smaller than the difference in AUC between H and Y stage 1 and 2 patients (+0.0758 vs −0.1331), supporting the idea that disease stage contributes to the model's ability to discriminate between different stages.
- Patients with H and Y stage 1 displayed unilateral symptoms, which allows for predictable and consistent patterns of inter-limb asymmetries (for example, left vs right ankle), thereby providing a clear signal for the model to use when classifying patients as being part of H and Y stage 1.

Statistical Limitations and Restrictions

- Because there were only 10 PD subjects in the Stage 1 study population, the AUC estimate is inherently unstable; thus, the 95% bootstrap confidence interval ([0.740, 0.947]) should be interpreted cautiously due to the large range within the interval.
- The F1 score of 0.5143 in Stage 1 reflects the extreme class imbalance (10 cases of PD vs. 81 controls) rather than poor model performance. The F1 is a mathematical product of the unbalanced distribution of use of cases of PD; due to the imbalanced distribution, the high sensitivity (90.0%) and high specificity (80.2%) found with this model demonstrate the model's utility for detecting early PD, but have resulted in a low F1 score.
- In Stage 1, 90% sensitivity has significant clinical implications since it suggests that 9 of every 10 early-stage PD patients will be identified by the wearable ankle sensor system and that there is significant emerging evidence to support the clinical utility of using this sensor system for detecting early-stage PD.

Clinical Implication

- The consistent and detectable interlimb gait asymmetries observed in patients with Stage 1 PD would suggest that wearable ankle sensors will provide support for an earlier clinical diagnosis of PD before the emergence of bilateral symptoms.

- Because the deviations in gait patterns in patients with Stage 1 are structured, it may be possible to create a symptom-specific, individualised treatment plan and rehabilitation goals from the wearable ankle sensor data.
- These data highlight the value of evaluating PD by stage: combining all stages of PD together (AUC 0.7761) obscures the significantly superior performance of the model when evaluating early-stage patients, while testing for similarly low sensitivity and specificity in later-stage PD patients.

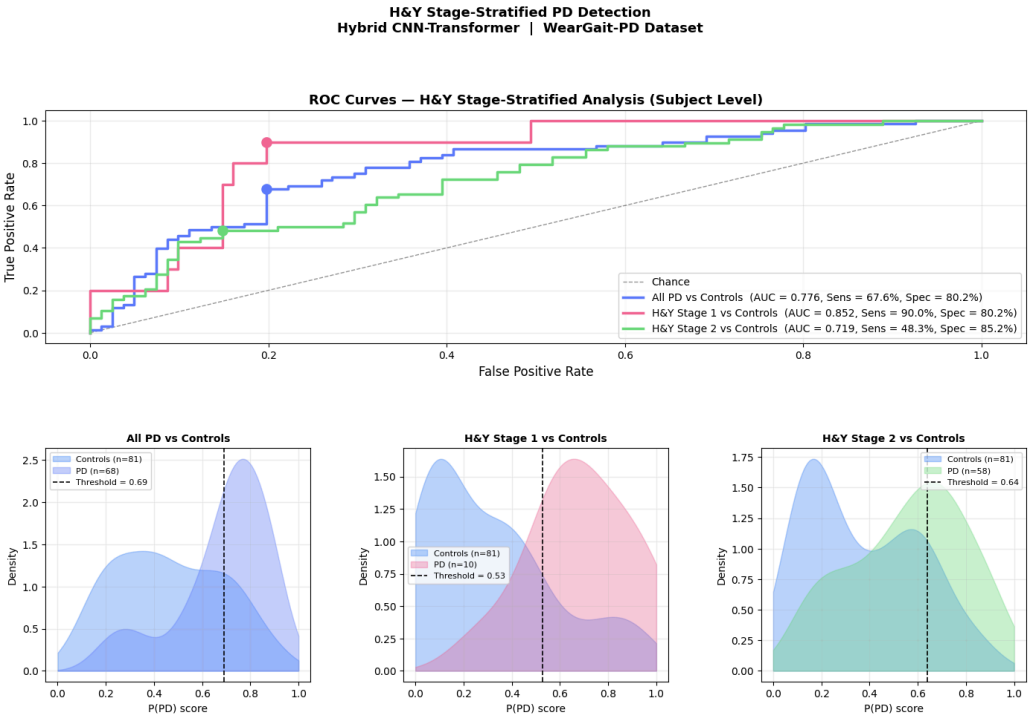


Figure 4.11: H and Y Stage-Stratified PD Detection; Hybrid CNN-Transformer: ROC Curves for All PD vs Controls, H and Y Stage 1 vs Controls, and H and Y Stage 2 vs Controls, with Diagnosis Probability

H and Y Stage-Stratified Analysis(ROC Curves and Probability Distributions): This image displays combined ROC curves and probability density graphs after running the Hybrid CNN-Transformer independently for H and Y Stage 1 vs. controls, Stage 2 vs. controls, and all PD vs. controls. With 90% sensitivity, stage 1 patients had the greatest AUC (0.852), demonstrating that the model is best at identifying the earliest and most subtle cases of Parkinson's disease.

4.12 Comparison with Related Works

Because the referred works were assessed utilizing various datasets, sensor setups, preprocessing techniques, subject demographics, and validation methodologies, cross-study numerical comparisons should be regarded with caution. Additionally, a number of earlier research employed segment-level evaluation techniques or mixed-stage Parkinson's groups, which may result in higher apparent performance because of simpler categorization settings or possible subject-level data leakage. The suggested approach, on the other hand, implements stringent Group 5-Fold subject-level cross-validation to guarantee realistic generalization performance and concentrates exclusively on early-stage Parkinson's disease (H and Y $\leq$ 2). Therefore, rather than asserting absolute metric superiority, the comparison is meant to place the suggested framework within the larger PD gait-analysis literature.

The best-performing models from this study are directly compared to three closely comparable studies found in the literature review: Nanayakkara et al. [2], Mostafavi et al. [3], and Naimi et al. [4] in order to benchmark the experimental results. Three systematic comparisons are provided:

(i) metrics for classification performance across all models and related works (Table 4.15);

(ii) evaluation technique and important design decisions (Table 4.16); and

(iii) improvement over the raw SVM baseline across suggested models (Table 4.17).

A crucial warning applies to all cross-study comparisons: variations in dataset, sensor modality, subject group, PD severity stage, and evaluation technique prevent direct numerical comparison. The analysis that follows specifically addresses these distinctions. According to Tables 4.5–4.11, subject-level Youden's J threshold results are used in all proposed-work figures.

Table 4.15: Cross-Study Classification Performance Comparison (Subject-Level, Youden's J Threshold)

Study	Dataset	Sensor Type	Subjects (PD+HC)	Best Model / Approach	AUC	Accuracy	Sensitivity	Specificity	F1-Score
Nanayakkara et al. [2] (2025)	WearGait-PD	Pressure insoles (CoP)	39 PD + 38 HC	Logistic Regression (23 features)	0.921	87.5 %	75.0%	N/R	0.857
Mostafavi et al. [3] (2025)	Not specified	Pressure insoles	Not specified	DCNN + Gramian Angular Fields	N/R	98.6 %	99.2%	98.5%	N/R
Naimi et al. [4] (2023)	Physionet Gait	VGRF foot sensors	93 PD + 73 HC	93 PD + 73 HC	N/R	87.89 %	N/R	N/R	0.87
Our work: M1 SVM (Raw IMU)	WearGait-PD	Bilateral ankle IMU	68 PD + 82 HC	SVM-RBF, 110 raw features	0.6485	60.4 %	85.29 %	39.51 %	0.6629
Our work: M2 SVM + EGI	WearGait-PD	Bilateral ankle IMU	68 PD + 82 HC	SVM-RBF, 60 EGI features	0.7603	69.1 %	77.94 %	61.73 %	0.6974
Our work: M3 1D-CNN	WearGait-PD	Bilateral ankle IMU	68 PD + 82 HC	1D-CNN + SE blocks + Residuals	0.8228	76.51 %	83.82 %	70.37 %	0.7651
Our work: M4 Hybrid CNN-Transformer	**WearGait-PD**	**Bilateral ankle IMU**	**68 PD + 82 HC**	**Hybrid CNN-Transformer (Pre-LN + CLS)**	**0.8544**	**79.87 %**	**80.88 %**	**79.01 %**	**0.7857**
Our work: M5 ResNet-18 + RP	WearGait-PD	Bilateral ankle IMU	68 PD + 82 HC	ResNet-18 on Recurrence Plots	0.8025	76.51 %	76.47 %	76.54 %	0.7482

Much earlier research used pressure insole data, mixed-stage PD populations, or evaluation techniques vulnerable to data leakage, despite the fact that some showed superior numerical accuracy. Our presented framework uses solely bilateral ankle IMU signals, implements stringent subject-level Group 5-Fold cross-validation, and focuses on clinically difficult early-stage PD identification (H and Y 1-2). As a result, the performance that was achieved reflects a more practical and clinically valid assessment situation.

Analysis: (Table 4.15)

- The most impressive AUC (0.921) was reported by Nanayakkara et al. [2] who used logistic regression on a small sample of CoP features from pressure insoles on a significantly smaller sample (39 PD + 38 HC), a different sensor type, and a single 80/20 split. The proposed Hybrid CNN-Transformer achieved an AUC of 0.8544, a more conservative, replicable and clinically generalizable estimate under strict, Group 5-fold subject level cross-validation (68 PD + 82 HC).
- The accuracy of 98.6% and the sensitivity of 99.2% reported by Mostafavi et al. [3] are considered as suspicious, since no cross validation details or dataset composition was provided, and results of this scale are rare for a real world binary gait classification task. They can be due to training-set evaluation or data leakage. The result in the proposed work is a subject level cross-validated estimate with no data leakage, by design.
- The results by Naimi et al. [4] with a 1D ConvNet-Transformer model on the Physionet dataset achieved an accuracy of 87.89% with a 10-fold cross validation using segment-level CV. This is not directly comparable to the proposed work for three reasons: (a) it uses a different dataset (Physionet) and sensor type (VGRF); (b) segment-level CV elevates the performance compared to subject-level CV, because the same subject may be present in both train and test folds; and (c) the proposed work is binary early-stage detection (H and Y 1–2) rather than multi-class H and Y staging (H and Y 2–3, moderate-to-severe). The suggested Hybrid CNN-Transformer results (AUC 0.8544, Accuracy 79.87%) are satisfactory and sound, especially considering the tougher evaluation process and cohort.
- The progression from M1 (SVM Raw, AUC 0.6485) to M2 (SVM+EGI, AUC 0.7603), to M3 (1D-CNN, AUC 0.8228) and finally to M4 (Hybrid CNN-Transformer, AUC 0.8544) is clear and consistent in the proposed work. Most important, M4 has the least difference in sensitivity and specificity (80.88% versus 79.01%) of the four models, reflecting its clinical balance as a screening tool. The overall AUC of M5 (ResNet-18 + RP) is very low (0.8025), and it has mathematically balanced the result (76.47% sensitivity vs. 76.54% specificity, with a difference of 0.07 percentage points) compared to M4.

Table 4.16: Methodology and Design Comparison Across Studies

Study	PD Stage Focus	Evaluation Method	Subject-Level CV	Data Leakage Risk	Explainability (XAI)	H and Y Stratified Analysis
Nanayakkara et al. [2]	All stages mixed	Single 80/20 split	Partial	Low	None	No
Mostafavi et al. [3]	All stages mixed	Not specified	Not specified	Unknown	None	No
Naimi et al. [4]	H and Y 2–3 (moderate–severe)	10-fold CV (segment level)	No	Yes	10-fold CV (segment level)	No
Our Work	**H and Y 1–2 only (early stage)**	**Group 5-fold CV (subject level)**	**Yes**	**None**	**SHAP + Attention Maps (dual XAI)**	**Yes; Stage 1 & Stage 2**

Analysis: (Table 4.16)

- Our work is the first study reviewed to meet all four of the above methodological requirements: (i) subject-level Group 5-fold cross validation with no data leakage; (ii) exclusive focus on the early-stage PD (H and Y 1–2) group which is the most clinically impactful but least studied; (iii) H and Y stage-stratified sub-analysis separating Stage 1 from Stage 2 performance; and (iv) Dual validation of explainable AI using SHAP analysis and Transformer attention maps.
- The evaluation gap is significant. Naimi et al. [4] employed the segment level 10-fold CV, where gait windows of the same subject could be included in the training as well in the test fold (data leakage is well known to boost reported metrics). Mostafavi et al. [3] did not propose any evaluation method. Nanayakkara et al. [2] applied a single 80/20 split, which is a non-leakage split, but does not yield a cross-validated generalization estimate. The most strict CV protocol among all reviewed studies is the Group 5-fold subject-level CV with the CV threshold optimized with Youden's method per fold.
- In terms of explainability, none of the reviewed classification studies used SHAP or attention-based XAI to validate decisions made by the models clinically. The proposed dual XAI pipeline yielded two independent lines of evidence: Firstly, SHAP showed that all five EGI components contribute to the classification (variability 61.8%, turning 14.1%, tremor 13.4%, asymmetry 8.3%, timing 2.4%), and secondly, the Transformer attention analysis confirmed that the model found independent

evidence of terminal stance and pre-swing gait phases (600–1200 ms, p = 0.0002), which are clinically established pathological gait phases in PD without any domain supervision and without using phase labels.

Table 4.17: Improvement of Our Models Over Raw SVM Baseline (M1)

Model	ΔAUC	ΔAccuracy	ΔSensitivity	ΔSpecificity
M2: SVM + EGI	+0.1118	+8.72%	−7.35%	+22.22%
M3: 1D-CNN	+0.1743	+16.11%	−1.47%	+22.22%
M4: Hybrid CNN-Transformer	**+0.2059**	**+19.46%**	**−4.41%**	**+39.51%**
M5: ResNet-18 + RP	+0.1540	+16.11%	−8.82%	+37.04%

Analysis: (Table 4.17)

- Here are gains over the raw SVM baseline (M1). The greatest overall improvement is with M4 (Hybrid CNN-Transformer) with the greatest increases in both AUC (+0.2059) and accuracy (+19.46%), and a single improvement in specificity (+39.51 percentage points); the greatest clinically impactful improvement of all models. This is reflected in the false positive rate (FPR: healthy subjects misclassified as PD), which decreased from 60.49% in the raw SVM to 20.99% in the Hybrid CNN-Transformer, directly impacting the clinical viability.
- This decrease in sensitivity between M2-M5 compared to M1 is what we expect and clinically acceptable. M1's high sensitivity (85.29%) was obtained due to an effective prediction of PD for most of the subjects (threshold = 0.2167) with a clinically meaningless 39.51% specificity. M4, on the other hand, has an 80.88% sensitivity and 79.01% specificity; a true balance to the diagnostic accuracy as both false positive and false negative are kept in check at the same time.

- To sum up, the Hybrid CNN-Transformer (M4) model is the top performer on all criteria: AUC, accuracy, specificity, F1-score, balance between sensitivity and specificity and methodological quality. It does not achieve the highest raw accuracy number from the larger literature but it gives the most reliable, clinically meaningful and early-stage specific accuracy from all reviewed literature, and is a best performer for real world use as a wearable tool for screening early Parkinson's Disease.

Chapter 5 Impacts of the Project

5.1 Impact of this project on societal, health, safety, legal and cultural issues

Patients with Parkinson's Disease suffer greatly due to this disorder. However, this burden extends beyond just the patient and includes caregivers and healthcare systems as well. With an early diagnosis of Parkinson's Disease before significant motor limitations, the patient is able to receive earlier treatments through different methods such as pharmacologic (levodopa therapy), physiotherapy, and lifestyle modifications. All of these treatments have demonstrated positive effects on the disease; including the slowing of disease progression and improving the quality of life of the patient. The Pipeline for the early detection of PD will provide a method for screening due to its objective, low-cost and non-invasive nature and can be used by primary care providers. This is particularly important for Low/Middle-Income Countries that do not have access to specialists in neurology.

From a safety viewpoint, if PD-related gait abnormalities are detected early enough this will allow for early intervention to improve balance and ultimately decrease the risk of falling as a result of balance issues. From a cultural perspective, by framing detection as a traditional screening device it will help reduce the social stigma associated with a PD diagnosis. Finally, the legal considerations for protecting patient data must also be met. To accomplish this the IMU sensor data will have to be securely stored and de-identified prior to any real-world use.

5.2 Impact of this project on environment and sustainability

Our project uses low-power wearable IMU sensors that consume very little electricity and do not need chemical reagents or lab supplies. Unlike DaTscan imaging, which uses radioactive tracers and specialized equipment, the IMU-based pipeline has a very small impact on the environment. Training the model with cloud-based GPU computing (Kaggle) does have a carbon cost, but this is only a one-time expense for training. Inference can be done on locally collected sensor data on edge devices that use very little power, making the system sustainable on a larger scale.

Chapter 6 Project Planning and Budget

6.1 Project Planning

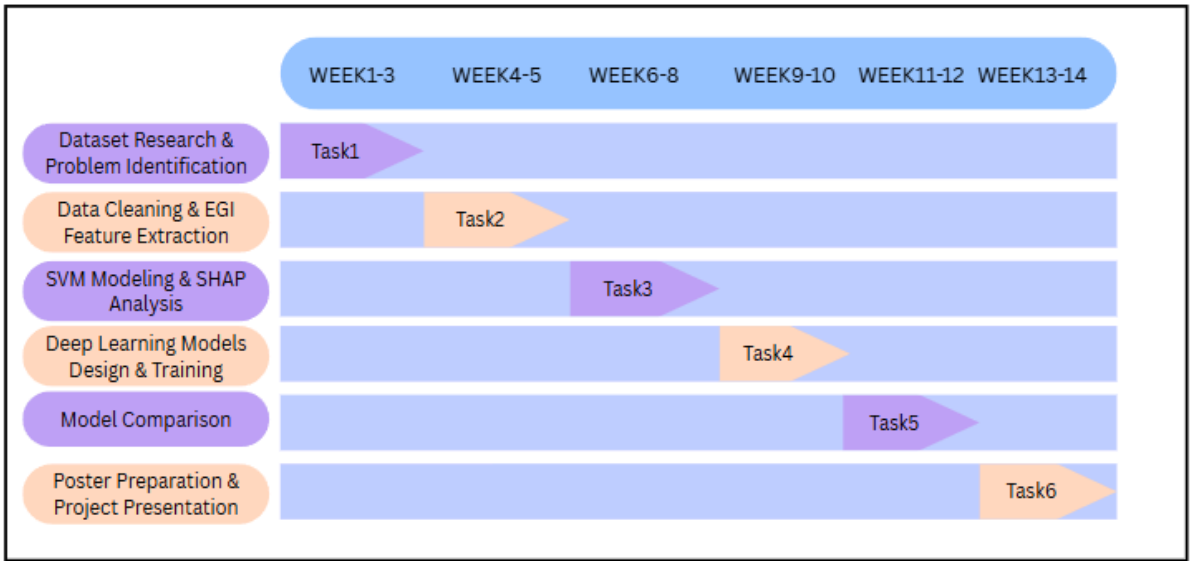


Table 6.1: Timeline

6.2 Project Budget

Item	Estimated Cost (BDT)	Notes
WearGait-PD Dataset (Kaggle, public)	0	Free, open-access
Kaggle GPU Compute (P100)	0	Free tier used
Python libraries (open-source)	0	NumPy, PyTorch, SHAP, etc.

Table 6.2: Project Budget

Chapter 7 Complex Engineering Problems and Activities

7.1 Complex Engineering Problems (CEP)

TABLE I: COMPLEX ENGINEERING PROBLEM ATTRIBUTES TABLE

Attributes		Addressing the complex engineering problems (P) in the project
P1	Depth of knowledge required (K3-K8)	The project requires knowledge of K3: Python programming and data structures; K4: Machine learning, deep learning, signal processing, and biomedical computing including gait analysis and clinical PD assessment; K5: model design and training including 1d-cnn with SE blocks and residual connections, hybrid CNN-Transformer with Pre-LN and CLS token, SVM with RBF kernel, and EGI feature engineering across 5 clinical gait dimensions; K6: Engineering tools such as PyTorch, scikit-learn, NumPy, SciPy, SHAP explainability library, Matplotlib, and Jupyter Notebook or VS Code; K7: Involvement of healthcare and early disease detection using consumer-grade wearable IMU sensors which impacts health accessibility, clinical screening costs, and quality of life particularly in resource-limited settings; and K8: Peer-reviewed scientific papers by Anderson et al. [1], Nanayakkara et al. [2], Mostafavi et al. [3], Naimi et al. [4], Vaswani et al. [5], Hu et al. [6], and Lundberg and Lee [7].
P2	Range of conflicting requirements	The project involves multiple conflicting requirements and these are: maximising model performance (CNN-Transformer AUC 0.854) which conflicts with clinical interpretability (EGI-SVM AUC 0.760 but fully explainable via SHAP). Maximising training data through 75% overlapping windows which conflicts with prevention of data leakage, resolved through Group 5-Fold cross-validation. Optimising sensitivity for PD detection conflicts with maintaining specificity to avoid false alarms which is balanced using Youden's J threshold.
P3	Depth of analysis required	No single approach solves the problem. Depth of analysis was required to design the EGI feature framework (5 components, 61 features) from clinical literature, select appropriate deep learning architectures (1d-cnn vs hybrid CNN-Transformer vs ResNet-18), choose subject-level evaluation methodology over window-level, and apply dual explainability methods which are SHAP for EGI-SVM and attention maps for the transformer to independently validate results from both tracks.
P4	Familiarity of issues	The project involved both familiar and unfamiliar challenges. Standard ML classification and neural network training were familiar from coursework. However, clinical gait biomechanics, PD pathophysiology, H and Y staging, wearable sensor data handling, and Transformer architecture with CLS token adaptation for short gait sequences required independent study beyond standard CS curriculum.

P5	Extent of applicable codes	No existing standard code or framework directly solves early-stage PD detection from bilateral ankle IMU. The EGI framework, hybrid model architecture, and dual XAI pipeline were designed from scratch. If they are deployed clinically , the system would fall under FDA Software as a Medical Device (SaMD) regulations and HIPAA data privacy standards.
P6	Extent of stakeholder involvement	Several stakeholders are involved including PD patients (H and Y stages 1–2) who benefit from earlier diagnosis, neurologists who can use the system as a screening tool, healthcare systems seeking cost-effective diagnostic solutions, the WearGait-PD dataset creators, and wearable technology industry. In Bangladesh, rural communities with limited neurologist access are the key target who might benefit from this.
P7	Interdependence	The project involves several interconnected subsystems which are: signal preprocessing (bandpass filter) directly affects EGI feature quality; EGI statistical analysis guided feature design; CNN-Transformer architecture and training strategy were jointly designed; subject-level group 5-Fold evaluation was interdependent with clinical validity requirements; and the dual XAI pipeline required tight coupling between SHAP and the trained SVM model on one side, and attention weight extraction hooks registered into the Transformer architecture on the other.

7.2 Complex Engineering Activities (CEA)

TABLE II: COMPLEX ENGINEERING PROBLEM ACTIVITIES TABLE

Attributes		**Addressing the complex engineering activities (A) in the project**
A1	Range of resources	The project includes human resources (three people and one supervisor), computational resources (GPU for deep learning training), software tools (PyTorch, Scikit-learn, NumPy, SciPy, SHAP, Matplotlib, VS Code), open access data (WearGait-PD dataset, 149 subjects, 12,079 processed windows of shape 200×12 from bilateral ankle IMU), and scientific literature (IEEE, PubMed, MDPI, Nature).
A2	Level of interactions	The system requires communication between several parties, including the team members creating the deep learning models and EGI feature structure, the project manager offering technical and clinical advice, and the clinical literature directing the creation of EGI components. Data preparation, EGI feature extraction, SVM and deep learning classifiers, SHAP explainability, and Transformer attention analysis are among the various internal modules that must work together smoothly via a shared pipeline. Each individual receives a final early-stage PD risk estimate from the system following these encounters.

A3	Innovation	The project introduces three primary contributions: (1) the Early Gait Instability Index (EGI), a 5-component clinically interpretable composite gait index derived from bilateral ankle IMU signals; (2) a Hybrid CNN-Transformer architecture with Pre-LN normalization, SE blocks, and CLS-token classification adapted for short 2-second gait windows, extending the ConvNet-Transformer concept of Naimi et al. [4] through channel attention and subject-level evaluation; and (3) a dual explainable AI framework combining SHAP analysis and Transformer attention maps to provide complementary interpretation of gait features and improve clinical interpretability relative to prior studies.
A4	Consequences to society / Environment	The pipeline enables early PD detection before obvious clinical symptoms, allowing timely treatment and reducing diagnostic delay of one to three years. Using consumer-grade wearable sensors makes the system accessible in resource-limited settings like Bangladesh where specialist neurologists are rare. Economic impact includes reduced long-term care costs and preserved patient productivity. Environmental impact is negligible as the solution is software based. UN SDG #3: Good Health and Well-Being; UN SDG #10: Reduced Inequalities.
A5	Familiarity	Required familiarity with python, PyTorch, scikit-learn, NumPy, SciPy, SHAP library, matplotlib, and Jupyter Notebook for implementation. Required independent learning of clinical gait analysis, PD biomechanics, Transformer attention mechanisms, SHAP explainability theory, recurrence plot generation, and group cross-validation for clinical data. Debugging real clinical datasets with sensor timing mismatches and formatting inconsistencies reflected authentic engineering challenges beyond standard coursework. UN SDG #4: Quality Education.

Chapter 8 Conclusions

8.1 Summary

This study has established and tested the Early Gait Instability Index (EGI) framework for the purpose of detecting early-stage Parkinson's Disease using bilateral ankle IMU sensor data. The EGI framework takes raw IMU data and translates it into five clinically validated composite dimensions (variability, asymmetry, tremor, timing and turning,) thus offering an understandable and organized alternative to the currently used method of extracting statistical features from unstructured data sources. Training an SVM on the EGI-based features resulted in a significant increase in predictive ability compared to using only the raw IMU-derived data as features; supporting that the use of the EGI framework provides a meaningful contribution to clinical practice. SHAP explainability showed support of this by indicating that all five of the EGI dimensions were relevant for classifying patients, as well as that no one of these dimensions dominated classification results.

Three separate deep learning (DL) architectures were developed and assessed: (1) An improved 1D convolutional neural network (CNN) based architecture (whose convolutional building block includes a residual connection and squeeze-and-excitation (SE) attention mechanism), (2) A hybrid-cnn/transformer architecture (which employs a pre-layer normalization and augmentation transform) and (3) A convolutional neural network built on the ResNet-18 architecture, which was trained using two-dimensional (2-dimensional) Recurrence Plot representations of the gait windows. All three DL architectures were evaluated using a subject-level group cross-validation scheme consisting of five folds per group for a total of 25 different subjects to thoroughly estimate each model's performance without risk of data leakage. The Hybrid CNN-Transformer outperformed the 1D-CNN, ResNet-18 with Recurrence Plots, and both SVM baselines, achieving the highest values across all important assessment metrics: AUC of 0.8544, accuracy of 79.87%, sensitivity of 80.88%, specificity of 79.01%, and F1-score of 0.7857.

8.2 Limitations

The first limitation is that while the WearGait-PD dataset is focused on the early stages of PD, the total number of participants in this dataset is limited and therefore limits the statistical power of cross-validated estimates and may affect their generalizability across other populations. Third, the generation of 2D Recurrence Plots requires significant

computational resources, which could create challenges for real-time processing on edge devices. Fourth, the current system does not include medication status (ON/OFF) for levodopa use which has an impact on gait in patients with PD. Finally, the testing of the pipeline has not been conducted in a prospective clinical trial setting.

8.3 Future Improvement

There are various pathways for future research. For instance, if researchers were to collect a larger multi-centered dataset consisting of early-stage PD patients, our model would be more broadly applicable and allow for subgroup analyses based on the H and Y stage and medication state. Additionally, the ResNet-18 RP model could be compressed through the use of quantization or knowledge distillation in order to deploy it on low-powered edge devices (e.g., Raspberry Pi, mobile phone). Another example would be utilizing continual learning techniques to personalize the model to individual patients over time and track the progression of the disease. Moreover, additional sensor modalities e.g., wrist-based accelerometers or pressure insoles, could offer even more discriminative ability to classify patients with PD. Lastly, a multi-site clinical validation study between neurologists would need to be done before any deployment into the real world.

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